

Random Reshuffling for Smooth Convex Optimization: Dominates Stochastic Gradient Descent*

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Abstract

Stochastic Gradient Descent (SGD) is one of the most classical optimization algorithms with favorable theoretical guarantees, yet its practical implementation differs subtly from its well-known form and is often referred to as Shuffling Stochastic Gradient Descent (Shuffling SGD). A particularly popular strategy in Shuffling SGD is Random Reshuffling (RR), which has achieved great empirical success. Despite its strong performance, RR has long been considered a heuristic due to a lack of theoretical support.

Over the last decade, people have finally established provable convergence rates for RR, thus justifying its observed superiority. However, for smooth convex optimization, two clouds over the convergence theory of RR remain. According to current theory, Shuffling SGD under RR converges only when the stepsize is smaller than a threshold proportional to $1/n$, where n is the number of summands in the objective (or the number of data points). Consequently, the optimally tuned theoretical rate of Shuffling SGD under RR is strictly worse than that of SGD when the number of epochs is smaller than another threshold proportional to n . These two restrictions heavily limit the applicability of existing theories and leave a critical mismatch with practice.

In this work, for the first time, we prove that RR dominates SGD in smooth convex optimization for any reasonable stepsize after any finite number of epochs, thereby addressing a longstanding open question. Moreover, we establish new algorithm-dependent lower bounds for RR and SGD, respectively. Altogether, these results yield (nearly) sharp worst-case characterizations of RR and SGD. Furthermore, we develop a shuffling-based method to optimize functions satisfying the quadratic growth condition, a strict relaxation of strong convexity, and derive a new complexity bound improving existing results proved under stronger assumptions.

1 Introduction

One of the fundamental tasks in machine learning is to optimize functions in a finite-sum form, i.e., $f(\mathbf{x}) \triangleq \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x})$. Among different optimization algorithms, Stochastic Gradient Descent (SGD), proposed in the seminal work of Robbins and Monro (1951), is arguably one of the most classical methods. Due to its easy implementation and computational efficiency, SGD is particularly popular when n is large, the standard case nowadays. More importantly, the convergence guarantees of SGD have been extensively studied, yielding provable rates in various settings (Polyak, 1987; Bottou et al., 2018; Lan, 2020), thereby providing a theoretical backbone for SGD.

However, compared with the standard form of SGD analyzed in theory, which uniformly samples a function to perform a gradient descent step at each iteration (also known as with-replacement sampling), the practical implementation differs subtly and is often referred to as Shuffling Stochastic Gradient Descent (Shuffling SGD). In Shuffling SGD, the optimization procedure is divided into K epochs, and within each epoch, the order in which functions are processed is determined by a permutation π of $\{1, \dots, n\}$ (also known as without-replacement sampling). A widely implemented strategy for generating π is Random Reshuffling (RR), which, in each epoch, independently and uniformly draws a new permutation from all possible ones.

Although Shuffling SGD under RR has achieved great empirical success across numerous experiments, it has long been considered a heuristic due to a lack of theoretical support. Over the last decade, beginning with

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the pioneering work of Gürbüzbalaban et al. (2021), people have finally established provable convergence rates for RR, thereby justifying its observed superiority over standard SGD.

In particular, for smooth convex optimization (i.e., each f_i is convex and L -smooth¹), RR with a constant stepsize² η is known to converge in expectation at the rate $\frac{D^2}{\eta n K} + \eta^2 n L \sigma_\star^2$ (e.g., Mishchenko et al. (2020); Nguyen et al. (2021)), provided that the stepsize satisfies $\eta \lesssim \frac{1}{nL}$, where D denotes the distance between the initial point and the optimal solution, and $\sigma_\star^2$ is the gradient variance at the minimizer. In comparison, SGD under the same setting guarantees an in-expectation convergence rate $\frac{D^2}{\eta n K} + \eta \sigma_\star^2$ (Garrigos and Gower, 2023), but only requires $\eta \leq \frac{1}{4L}$. Recently, two works (Cortild et al., 2025; Attia et al., 2025) have further extended the convergence theory for SGD to the regime $\eta < \frac{2}{L}$. To date, the best-known rate of SGD for any $\eta < \frac{2}{L}$ is $\frac{D^2}{(2-\eta L)\eta n K} + \frac{\eta \sigma_\star^2}{(2-\eta L)^2}$ (Attia et al., 2025). Clearly, RR converges faster than SGD in the regime $\eta \lesssim \frac{1}{nL}$, which has been recognized as theoretical evidence demonstrating the strong performance of RR.

Despite the progress discussed above, some important issues remain unaddressed. The most critical longstanding open question is that people still do not understand what happens to RR when η falls into the regime $\eta \gtrsim \frac{1}{nL}$. This point is critical because, given that n is typically large in modern tasks, the threshold $\frac{1}{nL}$ can be extremely small or even vanish, whereas stepsizes used in practice are usually at a constant level, thereby leaving a significant gap between theory and practice. More crucially, even if one temporarily assumes that the existing rate for RR mentioned earlier could be extended to allow $\eta < \frac{2}{L}$ (though no such theory has been established), it would still fail to explain the advantage of RR over SGD, as the term $\eta^2 n L \sigma_\star^2$ for RR is worse than the term $\eta \sigma_\star^2$ for SGD when $\eta \gtrsim \frac{1}{nL}$. This hints that an analysis different from existing ones may be needed.

Another issue implied by the above discussion is that the optimally tuned rate for RR induced by the existing bound is only $\frac{LD^2}{K} + (\frac{L\sigma_\star^2 D^4}{nK^2})^{\frac{1}{3}}$, which is better than the best tuned rate $\frac{LD^2}{nK} + \frac{\sigma_\star D}{\sqrt{nK}}$ for SGD only when K is larger than a threshold proportional to n . Moreover, in the case of $\sigma_\star = 0$ (i.e., all f_i 's share a common optimal solution), the rate of RR reduces to only $\frac{LD^2}{K}$, which is even worse than the $\frac{LD^2}{nK}$ rate of SGD by a factor of n .

The above restrictions on the stepsize η or the number of epochs K heavily limit the applicability of existing theories and cannot fully explain the favorable performance of Shuffling SGD under RR compared with standard SGD. Therefore, we are naturally led to the following question:

In smooth convex optimization, does RR dominate SGD without these two restrictions?

1.1 Our Contributions

This work provides an affirmative answer to the above question.

- Concretely, we show that Shuffling SGD under RR provably converges at a rate of $\frac{D^2}{(2-\eta L)\eta n K} + \frac{\min\{1, \eta n L\} \eta \sigma_\star^2}{(2-\eta L)^2}$ for any stepsize satisfying $\eta < \frac{2}{L}$ (see Theorem 1 for the formal version with nonuniform smoothness parameters and dynamic stepsizes that depend on the epoch number). We highlight that this rate is not only the first provable result for RR that allows $\eta \gtrsim \frac{1}{nL}$, but also provably dominates the bound $\frac{D^2}{(2-\eta L)\eta n K} + \frac{\eta \sigma_\star^2}{(2-\eta L)^2}$ of SGD under any stepsize $\eta < \frac{2}{L}$. It is noteworthy that this rate is not merely a simple extension of the previously best bound for RR, since the latter is slower than SGD when $\eta \gtrsim \frac{1}{nL}$, as discussed before. Therefore, for the first time, we prove that RR dominates SGD in smooth convex optimization for any reasonable stepsize after any finite number of epochs, resolving a longstanding open question. Consequently, the optimally tuned rate for Shuffling SGD under RR is improved to $\frac{LD^2}{nK} + \min\{\frac{\sigma_\star D}{\sqrt{nK}}, (\frac{L\sigma_\star^2 D^4}{nK^2})^{\frac{1}{3}}\}$ (see Corollary 1 for the formal version with nonuniform smoothness parameters), which dominates the best tuned bound $\frac{LD^2}{nK} + \frac{\sigma_\star D}{\sqrt{nK}}$ of SGD for any finite K . Moreover, the rate reduces to $\frac{LD^2}{nK}$ when $\sigma_\star = 0$ (i.e., the interpolation regime), improving upon the best known result $\frac{LD^2}{K}$ by a factor of $\frac{1}{n}$.

¹For simplicity, we adopt a uniform smoothness parameter in the discussion, as in most of the existing literature.

²We also use a constant stepsize in the discussion for convenience.

- In addition, we prove new algorithm-dependent lower bounds of $\frac{D^2}{\eta n K} + \min\{1, \eta n L\} \eta \sigma_\star^2$ for Shuffling SGD under RR in Theorem 2 and $\frac{D^2}{\eta n K} + \eta \sigma_\star^2$ for SGD in Theorem 3. Together with our new upper bound for RR and the existing upper bound for SGD, these results establish (nearly) sharp worst-case characterizations of the convergence of RR and SGD in the regime $\eta \leq \frac{1}{L}$ (and, more generally, $\eta \leq \frac{C}{L}$ for any absolute constant $C \in (0, 2)$).
- Furthermore, building on the classical restart technique in the optimization literature, we develop a restarted variant of Shuffling SGD termed Restarted Shuffling SGD. By applying the new upper bound for RR mentioned above, we show that Restarted Shuffling SGD under RR can be used to optimize functions satisfying the quadratic growth condition, which is strictly weaker than strong convexity. In particular, we establish a new complexity bound under this weaker assumption, improving upon the best known results for RR that require either strong convexity of f or strong convexity of each f_i . Moreover, our new bound is sharper than the corresponding bound for SGD across different parameter regimes under the same setting.

1.2 Related Work

We provide a brief overview of Shuffling SGD under RR and defer further details to Appendix A.

Over the past decades, the effectiveness of RR has been reported in many works (e.g., Bottou (2009, 2012); Bengio (2012)). However, the theoretical understanding of it has long lagged behind. The first breakthrough is by Gürbüzbalaban et al. (2021), which provides the first theoretical evidence that RR can beat SGD in smooth strongly convex optimization under certain additional assumptions. Since then, RR has been extensively studied (e.g., Nagaraj et al. (2019); Haochen and Sra (2019)). To date, the best known rate in smooth convex optimization is $\frac{D^2}{\eta n K} + \eta^2 n L \sigma_\star^2$ under the condition $\eta \lesssim \frac{1}{nL}$ (Mishchenko et al., 2020; Nguyen et al., 2021). The only existing lower bound in smooth convex optimization is $(\frac{L \sigma_\star^2 D^4}{n K^2})^{\frac{1}{3}}$, due to Cha et al. (2023), which holds for constant stepsizes $\eta \lesssim \frac{1}{nL}$ and a large number of epochs at least satisfying $K \gtrsim \frac{n L^2 D^2}{\sigma_\star^2}$.

2 Preliminaries

Notation. $\mathbb{N}$ denotes the set of natural numbers (excluding 0). Given $n \in \mathbb{N}$, we write $[n] \triangleq \{1, \dots, n\}$. $\lfloor \cdot \rfloor$ and $\lceil \cdot \rceil$ denote the floor and ceiling functions, respectively. $\langle \cdot, \cdot \rangle$ is the standard Euclidean inner product, and $\|\cdot\| \triangleq \sqrt{\langle \cdot, \cdot \rangle}$ is the ℓ_2 norm. Given a real-valued differentiable function $h : \mathbb{R}^d \rightarrow \mathbb{R}$, $\nabla h(\mathbf{x})$ denotes the gradient at $\mathbf{x} \in \mathbb{R}^d$. The Bregman divergence induced by h is defined as $B_h(\mathbf{x}, \mathbf{y}) \triangleq h(\mathbf{x}) - h(\mathbf{y}) - \langle \nabla h(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle$, which is nonnegative if h is additionally convex.

Objective. We study the following finite-sum optimization problem in this work

$$\inf_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}) \triangleq \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}),$$

where $n \in \mathbb{N}$ and each $f_i : \mathbb{R}^d \rightarrow \mathbb{R}$ is differentiable.

Remark 1. To ease notation, hereinafter we use $B \triangleq B_f$ and $B_i \triangleq B_{f_i}$ to denote the Bregman divergences induced by f and f_i , respectively.

Assumptions. Our analysis relies on the following three assumptions.

Assumption 1 (Minimizer). $\exists \mathbf{x}_\star \in \mathbb{R}^d$ such that $f_\star \triangleq f(\mathbf{x}_\star) = \inf_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}) \in \mathbb{R}$.

Assumption 2 (Convexity). Each f_i is convex.

Assumption 3 (Smoothness). Each f_i is L_i -smooth, i.e., $\exists L_i > 0$ such that $\|\nabla f_i(\mathbf{x}) - \nabla f_i(\mathbf{y})\| \leq L_i \|\mathbf{x} - \mathbf{y}\|$, $\forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^d$.

All three of the above assumptions are standard and commonly adopted in the literature (Polyak, 1987; Nesterov et al., 2018; Bottou et al., 2018; Lan, 2020). Notably, we do not impose any assumptions on the difference between the individual gradient ∇f_i and the full gradient ∇f , such as the popular finite variance condition.

Next, we introduce three more notations to simplify the expressions in the subsequent sections. $\sigma_\star^2$ denotes the variance of the gradient at the minimizer $\mathbf{x}_\star$, and $\bar{L}$ (resp. $\hat{L}$) represents the average (resp. maximum) smoothness parameter, i.e.,

$$\sigma_\star^2 \triangleq \frac{1}{n} \sum_{i=1}^n \|\nabla f_i(\mathbf{x}_\star)\|^2, \quad \bar{L} \triangleq \frac{1}{n} \sum_{i=1}^n L_i, \quad \hat{L} \triangleq \max_{i \in [n]} L_i.$$

We note that the quantity $\sigma_\star^2$ is widely used in prior works on shuffling gradient methods (e.g., Ying et al. (2019); Mishchenko et al. (2020); Nguyen et al. (2021)) and remains invariant even when f has multiple minimizers (see Lemma 4.17 of Garrigos and Gower (2023)). In particular, $\sigma_\star^2 = 0$ corresponds to the case in which all f_i share a common optimal solution.

To finish this section, we state a classical result in convex optimization, known as the co-coercivity property of smooth convex functions, which serves as a key tool in our analysis. As for its proof, see, for example, Theorem 2.15 of Nesterov et al. (2018).

Lemma 1 (Co-coercivity). *Let $h : \mathbb{R}^d \rightarrow \mathbb{R}$ be a differentiable convex function that is also L -smooth, then we have, for any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$,*

$$\|\nabla h(\mathbf{x}) - \nabla h(\mathbf{y})\|^2 \leq 2LB_h(\mathbf{x}, \mathbf{y}) \quad \text{and} \quad \|\nabla h(\mathbf{x}) - \nabla h(\mathbf{y})\|^2 \leq L \langle \nabla h(\mathbf{x}) - \nabla h(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle.$$

3 Shuffling Stochastic Gradient Descent

Algorithm 1 Shuffling Stochastic Gradient Descent (Shuffling SGD)

Input: initial point $\mathbf{x}_1^1 \in \mathbb{R}^d$, stepsize $\eta_k > 0$, number of epochs $K \in \mathbb{N}$
for $k = 1$ **to** K **do**
 Generate a permutation π_k of $[n]$
 for $i = 1$ **to** n **do**
 $\mathbf{x}_k^{i+1} = \mathbf{x}_k^i - \eta_k \nabla f_{\pi_k^i}(\mathbf{x}_k^i)$
 end for
 $\mathbf{x}_{k+1}^1 = \mathbf{x}_k^{n+1}$
end for

The method studied in this work, Shuffling Stochastic Gradient Descent (Shuffling SGD), is given in Algorithm 1. Compared with the standard SGD algorithm, which uniformly samples a function to process at each step, Shuffling SGD determines the order in which functions are passed in each epoch based on a permutation. In particular, three strategies for generating permutations are popular in practice, as illustrated in the following examples, where S_n denotes the symmetric group of $[n]$.

Example 1 (Random Reshuffling (RR)). Each π_k is drawn independently and uniformly from S_n .

Example 2 (Single Shuffling (SS)). Each $\pi_k = \pi$, a permutation drawn uniformly from S_n .

Example 3 (Incremental Gradient (IG)). Each $\pi_k = \pi$, a deterministic permutation from S_n .

3.1 New Rate for RR

We are now ready to provide the first main result, Theorem 1, a new convergence rate for RR.

Theorem 1. Under Assumptions 1, 2, and 3, suppose RR is employed with $\left\{\eta_k < \frac{2}{\bar{L}}\right\}_{k=1}^K$, let

$$\bar{\mathbf{x}}_K \triangleq \sum_{k=1}^K \sum_{i=1}^n \frac{(2 - \eta_k \hat{L})\eta_k}{nH_K} \mathbf{x}_k^i \quad \text{where} \quad H_K \triangleq \sum_{k=1}^K (2 - \eta_k \hat{L})\eta_k, \quad (1)$$

then *Shuffling SGD* (Algorithm 1) guarantees that

$$\mathbb{E}[f(\bar{\mathbf{x}}_K) - f_\star] \lesssim \frac{\|\mathbf{x}_1^1 - \mathbf{x}_\star\|^2}{n \sum_{k=1}^K (2 - \eta_k \hat{L})\eta_k} + \min \left\{ \frac{\sum_{k=1}^K \eta_k^2 / (2 - \eta_k \hat{L})}{\sum_{k=1}^K (2 - \eta_k \hat{L})\eta_k}, \frac{\sum_{k=1}^K \eta_k^3 n \bar{L} / (2 - \eta_k \hat{L})}{\sum_{k=1}^K (2 - \eta_k \hat{L})\eta_k} \right\} \sigma_\star^2.$$

Remark 2. As in the conference version (Liu, 2026), we can also consider the weighted average iterate $\bar{\mathbf{x}}_K \triangleq \sum_{k=1}^K \sum_{i=1}^n \frac{\eta_k}{nH_K} \mathbf{x}_k^i$, where $H_K \triangleq \sum_{k=1}^K \eta_k$, and prove a similar rate to that in Theorem 1, with $2 - \eta_k \hat{L}$ replaced by $2 - \hat{\eta} \hat{L}$, where $\hat{\eta} \triangleq \max_{k \in [K]} \eta_k$.

Proof. The proof is deferred to Subsection 4.4. $\square$

To the best of our knowledge, in smooth convex optimization, Theorem 1 shows the first convergence rate for RR under any reasonable stepsize, i.e., $\eta_k < 2/\hat{L}$, thereby improving existing results in different aspects. Previously, the best known convergence rate for RR is $\frac{D^2}{n \sum_{k=1}^K \eta_k} + \frac{\sum_{k=1}^K \eta_k^3 n \bar{L}}{\sum_{k=1}^K \eta_k} \sigma_\star^2$ (where $D \triangleq \|\mathbf{x}_1^1 - \mathbf{x}_\star\|$) under the condition $\eta_k \lesssim 1/(n\sqrt{\bar{L}\hat{L}})$ (Liu and Zhou, 2024; Cai et al., 2024). However, this requirement on η_k is highly unrealistic, as n is typically pretty large in modern machine learning tasks, meaning that the stepsize has to be extremely small or even vanishing. This contradicts the constant-level stepsizes commonly used in practice and leads to a gap between theory and practice.

In comparison, Theorem 1 not only allows the stepsize to lie in the constant regime $\eta_k < 2/\hat{L}$ but also shows a fundamental improvement rather than a mere extension of the previously best known rate, since the latter becomes slower than SGD once $\eta_k \gtrsim 1/(n\bar{L})$ (the rate of SGD is stated below), while our Theorem 1 never does.

In addition, compared with the rate $\frac{D^2}{n \sum_{k=1}^K (2 - \eta_k \hat{L})\eta_k} + \frac{\sum_{k=1}^K \eta_k^2 / (2 - \eta_k \hat{L})}{\sum_{k=1}^K (2 - \eta_k \hat{L})\eta_k} \sigma_\star^2$ of standard SGD (Attia et al., 2025)³, our Theorem 1 is never worse for any reasonable stepsize (i.e., $\eta_k < 2/\hat{L}$), and is strictly better when $\eta_k \lesssim 1/(n\bar{L})$. This feature has an important implication: if both methods employ their own optimally tuned stepsizes, Shuffling SGD under RR provably achieves a better upper bound that dominates the $\frac{\hat{L}D^2}{nK} + \frac{\sigma_\star D}{\sqrt{nK}}$ rate of SGD after any finite number of epochs, as evidenced by Corollary 1 below.

Corollary 1. Under the same setting as in Theorem 1, with the optimally tuned constant stepsize $\{\eta_k = \eta_\star\}_{k=1}^K$, where $\eta_\star < \frac{2}{\bar{L}}$, *Shuffling SGD* (Algorithm 1) guarantees that

$$\mathbb{E}[f(\bar{\mathbf{x}}_K) - f_\star] \lesssim \frac{\hat{L}\|\mathbf{x}_1^1 - \mathbf{x}_\star\|^2}{nK} + \min \left\{ \frac{\sigma_\star \|\mathbf{x}_1^1 - \mathbf{x}_\star\|}{\sqrt{nK}}, \left(\frac{\bar{L}\sigma_\star^2 \|\mathbf{x}_1^1 - \mathbf{x}_\star\|^4}{nK^2} \right)^{\frac{1}{3}} \right\}.$$

Proof. The proof is deferred to Subsection 4.4. $\square$

In contrast, the known optimally tuned rate for RR is only $\frac{\sqrt{\bar{L}\hat{L}}D^2}{K} + (\frac{\bar{L}\sigma_\star^2 D^4}{nK^2})^{\frac{1}{3}}$. To better understand the differences among these optimally tuned rates, let $\text{Order}_{\text{SGD}}(K)$ and $\text{Order}_{\text{RR}}^{\text{Old}}(K)$ denote the dominant terms in the existing optimally tuned rates for SGD and Shuffling SGD under RR, respectively, i.e.,

$$\text{Order}_{\text{SGD}}(K) \triangleq \max \left\{ \frac{\hat{L}D^2}{nK}, \frac{\sigma_\star D}{\sqrt{nK}} \right\} \quad \text{and} \quad \text{Order}_{\text{RR}}^{\text{Old}}(K) \triangleq \max \left\{ \frac{\sqrt{\bar{L}\hat{L}}D^2}{K}, \left(\frac{\bar{L}\sigma_\star^2 D^4}{nK^2} \right)^{\frac{1}{3}} \right\}.$$

³The work of Attia et al. (2025) does not provide a rate for dynamic stepsizes. However, their proof can be directly extended to this setting. For completeness, we include a separate theorem and prove it in Appendix F.

Similarly, $\text{Order}_{\text{RR}}^{\text{New}}(K)$ denotes the dominant term in the rate obtained in Corollary 1, i.e.,

$$\text{Order}_{\text{RR}}^{\text{New}}(K) \triangleq \max \left\{ \frac{\hat{L}D^2}{nK}, \min \left\{ \frac{\sigma_* D}{\sqrt{nK}}, \left(\frac{\bar{L}\sigma_*^2 D^4}{nK^2} \right)^{\frac{1}{3}} \right\} \right\}.$$

In the nondegenerate case $\sigma_* \neq 0$ (which implies that $n \geq 2$), the following comparisons hold:

$$\begin{cases} \text{Order}_{\text{RR}}^{\text{New}}(K) = \text{Orders}_{\text{SGD}}(K) \stackrel{(a)}{\leq} \text{Order}_{\text{RR}}^{\text{Old}}(K), & K \leq \frac{n\bar{L}^2 D^2}{\sigma_*^2}, \\ \text{Order}_{\text{RR}}^{\text{New}}(K) < \text{Orders}_{\text{SGD}}(K) \stackrel{(b)}{\leq} \text{Order}_{\text{RR}}^{\text{Old}}(K), & \frac{n\bar{L}^2 D^2}{\sigma_*^2} < K \leq \frac{n\bar{L}\hat{L}D^2}{\sigma_*^2}, \\ \text{Order}_{\text{RR}}^{\text{New}}(K) \stackrel{(c)}{\leq} \text{Order}_{\text{RR}}^{\text{Old}}(K) < \text{Orders}_{\text{SGD}}(K), & \frac{n\bar{L}\hat{L}D^2}{\sigma_*^2} < K \leq \frac{n\bar{L}^{1/2}\hat{L}^{3/2}D^2}{\sigma_*^2}, \\ \text{Order}_{\text{RR}}^{\text{New}}(K) = \text{Order}_{\text{RR}}^{\text{Old}}(K) < \text{Orders}_{\text{SGD}}(K), & K > \frac{n\bar{L}^{1/2}\hat{L}^{3/2}D^2}{\sigma_*^2}, \end{cases}$$

where (a) becomes an equality if and only if $K = \frac{n\bar{L}^2 D^2}{\sigma_*^2}$ and $\bar{L} = \hat{L}$, (b) becomes an equality if and only if $K = \frac{n\bar{L}\hat{L}D^2}{\sigma_*^2}$, and (c) becomes an equality if and only if $K = \frac{n\bar{L}^{1/2}\hat{L}^{3/2}D^2}{\sigma_*^2}$. As one can see, unlike the existing optimally tuned bound for RR in the literature, which can be slower than standard SGD when $K \leq \frac{n\bar{L}\hat{L}D^2}{\sigma_*^2}$, the new result in Corollary 1 is never worse than the optimally tuned rate of standard SGD and strictly beats it once $K > \frac{n\bar{L}^2 D^2}{\sigma_*^2}$, thereby improving the threshold from $\frac{n\bar{L}^2 D^2}{\sigma_*^2}$ to $\frac{n\bar{L}\hat{L}D^2}{\sigma_*^2}$.

Moreover, in the special case where all f_i 's share a common minimizer, or equivalently when $\sigma_* = 0$, Corollary 1 achieves the same $\frac{\hat{L}D^2}{nK}$ rate as SGD. However, the prior bound for RR can only be reduced to a slower rate in the order of $\frac{\sqrt{\bar{L}\hat{L}}D^2}{K}$ due to the stepsize restriction $\eta_k \lesssim 1/(n\sqrt{\bar{L}\hat{L}})$, as discussed before. Formally, once $n \geq 2$, we have

$$\text{Order}_{\text{RR}}^{\text{New}}(K) = \text{Orders}_{\text{SGD}}(K) < \text{Order}_{\text{RR}}^{\text{Old}}(K), \forall K \in \mathbb{N}.$$

In summary, Theorem 1 and Corollary 1 together imply that RR dominates SGD in smooth convex optimization under any reasonable stepsize after any finite number of epochs.

3.2 New Lower Bounds for RR and SGD

The careful reader may notice that our previous comparison focuses only on upper bounds, which alone may not fully substantiate the claim that RR dominates SGD. Motivated by this, we establish new algorithm-dependent lower bounds for both RR and standard SGD under a constant stepsize $\{\eta_k = \eta\}_{k=1}^K$.

Before stating these lower bounds, we introduce some notation for clarity. Given $d \in \mathbb{N}$, $n \geq 2$, $D > 0$, $L_{1:n} \triangleq \{L_i > 0\}_{i=1}^n$, and $\sigma_* \geq 0$, we let

$$\mathcal{F}(d, n, D, L_{1:n}, \sigma_*) \triangleq \left\{ f = \frac{1}{n} \sum_{i=1}^n f_i : \begin{array}{l} \exists \mathbf{x}_* \in \mathbb{R}^d \text{ such that } f_* \triangleq f(\mathbf{x}_*) = \inf_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}) \text{ and } \|\mathbf{x}_*\| \leq D, \\ \text{each } f_i : \mathbb{R}^d \rightarrow \mathbb{R} \text{ is differentiable, convex, and } L_i\text{-smooth,} \\ \frac{1}{n} \sum_{i=1}^n \|\nabla f_i(\mathbf{x}_*)\|^2 \leq \sigma_*^2. \end{array} \right\}.$$

In words, the class $\mathcal{F}(d, n, D, L_{1:n}, \sigma_*)$ contains all finite-sum objectives satisfying the required assumptions.

We are now ready to present the new lower bounds. The first is Theorem 2, a new lower bound for Shuffling SGD with RR.

Theorem 2. *Given $d \in \mathbb{N}$, $n \geq 2$, $D > 0$, $\{L_i > 0\}_{i=1}^n$, and $\sigma_* \geq 0$, suppose RR is employed with $\{\eta_k = \eta\}_{k=1}^K$ for $\eta > 0$, let $\bar{\mathbf{x}}_K \triangleq \frac{1}{nK} \sum_{k=1}^K \sum_{i=1}^n \mathbf{x}_k^i$ denote the average iterate generated by Shuffling SGD (Algorithm 1), initialized at $\mathbf{x}_1^1 = \mathbf{0}$, over $K \in \mathbb{N}$ epochs, then we have*

$$\sup_{f \in \mathcal{F}(d, n, D, L_{1:n}, \sigma_*)} \mathbb{E}[f(\bar{\mathbf{x}}_K) - f_*] \gtrsim \frac{D^2}{nK} + \min\{1, \eta n \bar{L}\} \eta \sigma_*^2 \quad \text{when } K \eta n \bar{L} \geq 1.$$

Remark 3. Although the theorem is stated under the condition $K\eta n\bar{L} \geq 1$, our proof applies to any $\eta > 0$ and $K \in \mathbb{N}$. We restrict our attention to the regime $K\eta n\bar{L} \geq 1$, since our analysis indicates that, in the regime $K\eta n\bar{L} < 1$, $\sup_{f \in \mathcal{F}(d, n, D, L_{1:n}, \sigma_*)} \mathbb{E}[f(\bar{\mathbf{x}}_K) - f_*] \gtrsim \bar{L}D^2$, suggesting that no meaningful convergence occurs for RR in this situation.

Proof. The proof is deferred to Subsection 5.3. $\square$

Theorem 2 provides a new algorithm-dependent lower bound for RR, extending the existing lower bound $(\frac{L\sigma_*^2 D^4}{nK^2})^{\frac{1}{3}}$ for $\{L_i = L\}_{i=1}^n$ under the conditions $\eta \lesssim 1/(nL)$ and $K \gtrsim nL^2 D^2 / \sigma_*^2$ (Cha et al., 2023) to every prespecified $\eta > 0$ and $K \in \mathbb{N}$ for nonuniform $\{L_i\}_{i=1}^n$ satisfying $K\eta n\bar{L} \geq 1$, the meaningful regime discussed in Remark 3. Importantly, this allows a direct comparison with stepsize-dependent upper bounds and provides a finer characterization of the limitations of RR. As one can observe, there remains a gap between the upper and lower bounds by factors polynomial in $1/(2 - \eta\hat{L})$ when $\eta < 2/\hat{L}$, and resolving this discrepancy is left for future work.

We highlight that Theorem 2 accommodates nonuniform smoothness parameters, therefore requiring a more careful analysis. At a high level, the analysis is divided into two parts. In the first part, we establish the term $\frac{D^2}{\eta n K}$ to capture the error induced by the initial distance between $\mathbf{x}_1^1$ and $\mathbf{x}_*$. In the second part, inspired by Rajput et al. (2020); Yun et al. (2022); Cha et al. (2023), we also construct asymmetric piecewise quadratic functions that allow us to explicitly track the RR iterates and establish the second term $\min\{1, \eta n\bar{L}\} \eta \sigma_*^2$, which captures the lower bound arising from σ_*^2 . For more details, please refer to Section 5.

Theorem 3. Given $d \in \mathbb{N}$, $n \geq 2$, $D > 0$, $\{L_i > 0\}_{i=1}^n$, and $\sigma_* \geq 0$, suppose a constant stepsize $\{\eta_k = \eta\}_{k=1}^K$ for $\eta > 0$ is employed, let $\bar{\mathbf{x}}_K \triangleq \frac{1}{nK} \sum_{k=1}^K \sum_{i=1}^n \mathbf{x}_k^i$ denote the average iterate generated by SGD (Algorithm 1, with π_k^i sampled independently and uniformly from $[n]$), initialized at $\mathbf{x}_1^1 = \mathbf{0}$, over $K \in \mathbb{N}$ epochs, then we have

$$\sup_{f \in \mathcal{F}(d, n, D, L_{1:n}, \sigma_*)} \mathbb{E}[f(\bar{\mathbf{x}}_K) - f_*] \gtrsim \frac{D^2}{\eta n K} + \eta \sigma_*^2 \quad \text{when } K\eta n\bar{L} \geq 1.$$

Remark 4. Similar to Theorem 2, we focus on the regime $K\eta n\bar{L} \geq 1$, since our proof suggests that, for SGD as well, $\sup_{f \in \mathcal{F}(d, n, D, L_{1:n}, \sigma_*)} \mathbb{E}[f(\bar{\mathbf{x}}_K) - f_*] \gtrsim \bar{L}D^2$ when $K\eta n\bar{L} < 1$.

Proof. The proof is deferred to Section 6. $\square$

Theorem 3 establishes a new algorithm-dependent lower bound for SGD, which, again, holds for every prespecified $\eta > 0$ and $K \in \mathbb{N}$ satisfying $K\eta n\bar{L} \geq 1$. As in the case of RR, the lower bound in Theorem 3 also differs from the upper bound in Theorem 11 by factors polynomial in $1/(2 - \eta\hat{L})$ when $\eta < 2/\hat{L}$, and closing this gap remains open.

The proof of Theorem 3 is relatively simpler than that of Theorem 2 due to the additional independence in the sampling process. In particular, leveraging this independence, we can analyze the iterates of SGD via a direct recursion and evaluate it in expectation, without having to track the dependence induced by sampling without replacement. For more details, we refer the interested reader to Section 6.

3.3 Worst-Case Characterization

Combining the new upper and lower bounds, we obtain (nearly) sharp worst-case characterizations for both RR and SGD with a constant stepsize η in smooth convex optimization. Concretely, given $\text{Alg} \in \{\text{RR}, \text{SGD}\}$, $K \in \mathbb{N}$, $\eta \in (0, 1/\hat{L}]$, and $f \in \mathcal{F}(d, n, D, L_{1:n}, \sigma_*)$, where $d \in \mathbb{N}$, $n \geq 2$, $D > 0$, $\{L_i > 0\}_{i=1}^n$, $\sigma_* \geq 0$, we define

$$\text{Rate}_{\text{Alg}}^{\text{Avg}}(K, \eta, f) \triangleq f(\bar{\mathbf{x}}_K) - f_*,$$

where, as before, $\bar{\mathbf{x}}_K = \frac{1}{nK} \sum_{k=1}^K \sum_{i=1}^n \mathbf{x}_k^i$ denotes the average iterate generated by Alg, initialized at $\mathbf{x}_1^1 = \mathbf{0}$, over K epochs.

Remark 5. For simplicity, we restrict our attention to the regime $\eta \in (0, 1/\hat{L}]$ to drop the term $1/(2 - \eta\hat{L})$ in the upper bounds for RR (see Theorem 1) and SGD (see Theorem 11).

With this notation, Theorems 1 and 2 for RR, together with Theorems 3 and 11 for SGD, give

$$\sup_{f \in \mathcal{F}(d, n, D, L_{1:n}, \sigma_*)} \mathbb{E} \left[\text{Rate}_{\text{Alg}}^{\text{Avg}}(K, \eta, f) \right] \asymp \text{Rate}_{\text{Alg}}^{\text{Avg}, \mathbb{E}}(K, \eta), \forall K \in \mathbb{N}, \eta \in (0, 1/\hat{L}], K\eta n\bar{L} \geq 1, \quad (2)$$

where

$$\text{Rate}_{\text{Alg}}^{\text{Avg}, \mathbb{E}}(K, \eta) \triangleq \begin{cases} \frac{D^2}{\eta n K} + \min\{1, \eta n \bar{L}\} \eta \sigma_*^2, & \text{Alg} = \text{RR}, \\ \frac{D^2}{\eta n K} + \eta \sigma_*^2, & \text{Alg} = \text{SGD}. \end{cases}$$

(2) immediately yields that, under the stated conditions,

$$\sup_{f \in \mathcal{F}(d, n, D, L_{1:n}, \sigma_*)} \mathbb{E} \left[\text{Rate}_{\text{RR}}^{\text{Avg}}(K, \eta, f) \right] \lesssim \sup_{f \in \mathcal{F}(d, n, D, L_{1:n}, \sigma_*)} \mathbb{E} \left[\text{Rate}_{\text{SGD}}^{\text{Avg}}(K, \eta, f) \right].$$

To the best of our knowledge, Theorem 7 of Haochen and Sra (2019)⁴ is the only analogous worst-case dominance result in the literature. Their result considers a different setting in which each f_i is μ_i -strongly convex and L_i -smooth, all f_i 's share a common minimizer (equivalently, $\sigma_* = 0$), and the stepsize satisfies $\eta \in (0, \min_{i \in [n]} 2/(L_i + \mu_i)]$. Under these conditions, their theorem states that the worst-case squared-distance error of RR is no larger than that of SGD after any number of iterations.

Discussion of De Sa (2020). Readers familiar with the literature of RR may notice that, at first glance, our result appears to contradict the work of De Sa (2020), which shows that RR is not uniformly better than SGD. We note that this apparent discrepancy arises from the use of different convergence measurements. Similarly, for $\text{Alg} \in \{\text{RR}, \text{SGD}\}$, $K \in \mathbb{N}$, $\eta \in (0, 1/\hat{L}]$, and $f \in \mathcal{F}(d, n, D, L_{1:n}, \sigma_*)$, we introduce

$$\text{Rate}_{\text{Alg}}^{\text{Last}}(K, \eta, f) \triangleq f(\mathbf{x}_K^{n+1}) - f_*$$

Under this notation, the finding of De Sa (2020) should be understood as an instance-dependent result. Concretely, Section 5 of De Sa (2020) constructs a function $\tilde{f} \in \mathcal{F}(d = 3, n = 4, D, \{L_1 = (2 + \sqrt{3})/2, L_2 = 1 + \cos(\pi/7), L_3 = L_4 = 2\}, \sigma_* = 0)$, where $D > 0$ can be arbitrary, such that

$$\Pr \left[\text{Rate}_{\text{RR}}^{\text{Last}}(K, 1/\hat{L}, \tilde{f}) > 0, \forall K \in \mathbb{N} \right] = 1 \quad \text{and} \quad \Pr \left[\text{Rate}_{\text{SGD}}^{\text{Last}}(K, 1/\hat{L}, \tilde{f}) = 0 \text{ eventually} \right] = 1.$$

We emphasize two subtle points here. First, the above result is established for the last iterate rather than the average iterate considered above. Second, and more importantly, this almost-sure convergence dominance does not necessarily imply, even for the constructed $\tilde{f}$, either

$$\mathbb{E} \left[\text{Rate}_{\text{SGD}}^{\text{Last}}(K, 1/\hat{L}, \tilde{f}) \right] < \mathbb{E} \left[\text{Rate}_{\text{RR}}^{\text{Last}}(K, 1/\hat{L}, \tilde{f}) \right] \quad \text{or} \quad \mathbb{E} \left[\text{Rate}_{\text{SGD}}^{\text{Avg}}(K, 1/\hat{L}, \tilde{f}) \right] < \mathbb{E} \left[\text{Rate}_{\text{RR}}^{\text{Avg}}(K, 1/\hat{L}, \tilde{f}) \right].$$

Therefore, the conclusion of De Sa (2020) does not contradict ours, as the two results concern fundamentally different settings: worst-case comparison versus instance-dependent comparison.

3.4 Application to Functions with Quadratic Growth

In this part, we provide an application of our new Theorem 1 to functions satisfying the quadratic growth condition in Assumption 4 below.

Assumption 4 (Quadratic growth). *f satisfies the μ -quadratic growth condition, i.e., $f(\mathbf{x}) - f_* \geq \frac{\mu}{2} \text{dist}^2(\mathbf{x}, \mathbb{X}_*)$, $\forall \mathbf{x} \in \mathbb{R}^d$ for some $\mu > 0$, where $\mathbb{X}_* \triangleq \arg \min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x})$ and $\text{dist}(\mathbf{x}, \mathbb{X}_*) \triangleq \inf_{\mathbf{x}_* \in \mathbb{X}_*} \|\mathbf{x} - \mathbf{x}_*\|$.*

The quadratic growth condition has a long history in the optimization literature (Ioffe, 1994; Bonnans and Ioffe, 1995; Anitescu, 2000; Necoara et al., 2019). It is a strict relaxation of strong convexity, i.e., every strongly convex function satisfies the quadratic growth condition, whereas the converse does not hold. It is also closely related to several other well-known conditions, including the Polyak-Łojasiewicz inequality (Polyak, 1963), error bounds (Luo and Tseng, 1993), and the restricted secant inequality (Zhang and Yin,

⁴The theorem numbering follows the arXiv version of Haochen and Sra (2019).

2013). We refer the interested reader to, e.g., Karimi et al. (2016) and Guille-Escuret et al. (2021) for systematic studies of these relationships.

Algorithm 2 Restarted Shuffling Stochastic Gradient Descent (Restarted Shuffling SGD)

Input: initial point $\mathbf{x}_1 \in \mathbb{R}^d$, stepsize $\eta > 0$, number of epochs $K \in \mathbb{N}$, number of restarts $T \in \mathbb{N}$

for $t = 1$ **to** T **do**

$\mathbf{x}_{t+1} = \text{Shuffling SGD}(\mathbf{x}_t, \eta, K)$, where $\mathbf{x}_{t+1}$ is constructed based on (1)

end for

To handle Assumption 4, inspired by the classical restart technique in the literature, e.g., Nesterov (1983), we consider a restarted variant of Shuffling SGD, termed **Restarted Shuffling SGD**, as presented in Algorithm 2. Specifically, we use **Shuffling SGD** as a subroutine and restart it after several epochs. At each restart, the output of the previous stage is used as the initialization for the next run. Theorem 4 below provides a general convergence guarantee for this restarted scheme.

Theorem 4. *Under Assumptions 1, 2, 3, and 4, suppose RR is employed by Shuffling SGD with $\eta < \frac{2}{\hat{L}}$ and $K \asymp K_\star \triangleq \frac{1}{(2-\eta\hat{L})\eta n\mu}$, then Restarted Shuffling SGD (Algorithm 2) guarantees that*

$$\mathbb{E}[f(\mathbf{x}_{T+1}) - f_\star] \lesssim \frac{\mu \text{dist}^2(\mathbf{x}_1, \mathbb{X}_\star)}{2^T} + \frac{\min\{1, \eta n \bar{L}\} \eta \sigma_\star^2}{(2 - \eta \hat{L})^2}.$$

Proof. The proof is deferred to Subsection 7.1. □

Theorem 4 shows that, with a constant stepsize η and a suitable choice of the number of epochs K between restarts, **Restarted Shuffling SGD** with RR reduces the function value gap at a linear rate up to an additive stochastic error. In particular, when $\sigma_\star = 0$, i.e., the interpolation regime, the function value gap $\mathbb{E}[f(\mathbf{x}_{T+1}) - f_\star]$ converges linearly to zero, and consequently so does $\mathbb{E}[\text{dist}^2(\mathbf{x}_{T+1}, \mathbb{X}_\star)]$.

At a high level, the intuition behind Theorem 4 is as follows. At each restart stage, Theorem 1 allows us to upper bound $\mathbb{E}[f(\mathbf{x}_{t+1}) - f_\star]$ in terms of $\mathbb{E}[\text{dist}^2(\mathbf{x}_t, \mathbb{X}_\star)]$ up to an additive error. The quadratic growth condition, in turn, lower bounds $f(\mathbf{x}_{t+1}) - f_\star$ by $\text{dist}^2(\mathbf{x}_{t+1}, \mathbb{X}_\star)$. With an appropriate choice of the restart period, these two inequalities yield a constant-factor contraction of $\mathbb{E}[\text{dist}^2(\mathbf{x}_t, \mathbb{X}_\star)]$ at each stage. Repeating this argument leads to the geometric decay stated in Theorem 4, while keeping the stochastic error at the same order as in Theorem 1.

Corollary 2. *Under the same setting as in Theorem 4, given $0 < \varepsilon < \mu \text{dist}^2(\mathbf{x}_1, \mathbb{X}_\star)$ sufficiently small, with $\eta = \eta_\star \triangleq \min\left\{\frac{1}{\hat{L}}, \max\left\{\frac{\varepsilon}{\sigma_\star^2}, \frac{1}{\sigma_\star} \sqrt{\frac{\varepsilon}{n\bar{L}}}\right\}\right\}$ and $T = T_\star \triangleq \left\lceil \log_2 \frac{\mu \text{dist}^2(\mathbf{x}_1, \mathbb{X}_\star)}{\varepsilon} \right\rceil$, Restarted Shuffling SGD (Algorithm 2) guarantees that*

$$\mathbb{E}[f(\mathbf{x}_{T_\star+1}) - f_\star] \lesssim \varepsilon,$$

using the following number of individual gradient evaluations:

$$\left(\frac{\hat{L}}{\mu} + \min\left\{ \frac{\sigma_\star^2}{\mu \varepsilon}, \frac{\sigma_\star}{\mu} \sqrt{\frac{n\bar{L}}{\varepsilon}} \right\} \right) \log \frac{\mu \text{dist}^2(\mathbf{x}_1, \mathbb{X}_\star)}{\varepsilon}.$$

Proof. The proof is deferred to Subsection 7.2. □

With Theorem 4 in hand, we further establish Corollary 2, which provides the number of individual gradient evaluations required to reach an ε -accurate solution. To the best of our knowledge, when each f_i is smooth and convex, our sample complexity bound is the sharpest currently available among shuffling-based methods, improving upon the best known results obtained under assumptions strictly stronger than the quadratic growth condition: $\left(\frac{n\sqrt{\bar{L}\hat{L}}}{\mu} A + \frac{\sigma_\star}{\mu} \sqrt{\frac{n\bar{L}}{\varepsilon}} B \right) \log \left(\frac{\mu \text{dist}^2(\mathbf{x}_1, \mathbb{X}_\star)}{\varepsilon} \right)$ for μ -strongly convex f (Liu and

Zhou, 2024), where $A \asymp \sqrt{\log \left(\frac{\sqrt{\bar{L}\hat{L}}}{\mu} \log \left(\frac{\mu \text{dist}^2(\mathbf{x}_1, \mathbb{X}_\star)}{\varepsilon} \right) \right)}$ and $B \asymp \sqrt{\log \left(\frac{\sigma_\star}{\mu} \sqrt{\frac{\bar{L}}{n\varepsilon}} \log \left(\frac{\mu \text{dist}^2(\mathbf{x}_1, \mathbb{X}_\star)}{\varepsilon} \right) \right)}$, and

$\left(\frac{\hat{L}}{\mu} + \frac{\sigma_*}{\mu} \sqrt{\frac{n\hat{L}}{\varepsilon}}\right) \log \frac{\mu \text{dist}^2(\mathbf{x}_1, \mathbb{X}_*)}{\varepsilon}$ for μ -strongly convex f_i (Mishchenko et al., 2020), both of which are achieved by **Shuffling SGD** with **RR**.

We would like to emphasize that, although our bound has a better order, it is achieved through a restarted scheme, namely **Restarted Shuffling SGD**. In contrast, the results of Liu and Zhou (2024) and Mishchenko et al. (2020) mentioned above do not require restarts. Therefore, an interesting direction for future work is to determine whether **Shuffling SGD** without restarts can achieve a similar guarantee for functions satisfying the quadratic growth condition.

For **SGD**, as far as we know, the best sample complexity bound in the literature that applies to our setting is $\frac{\hat{L}}{\mu} \log \frac{\bar{L}(f(\mathbf{x}_1) - f_*)}{\mu\varepsilon} + \frac{\hat{L}\sigma_*^2}{\mu^2\varepsilon}$, as implied by Corollary 3 of Khaled and Richtárik (2023). Since $\mu \text{dist}^2(\mathbf{x}_1, \mathbb{X}_*) \lesssim f(\mathbf{x}_1) - f_*$ and $\mu \leq \bar{L}$, their first term $\frac{\hat{L}}{\mu} \log \frac{\bar{L}(f(\mathbf{x}_1) - f_*)}{\mu\varepsilon}$ is hence worse than the corresponding term $\frac{\hat{L}}{\mu} \log \frac{\mu \text{dist}^2(\mathbf{x}_1, \mathbb{X}_*)}{\varepsilon}$ in our bound. Although their second term $\frac{\hat{L}\sigma_*^2}{\mu^2\varepsilon}$ is not directly comparable in all parameter regimes with our $\min \left\{ \frac{\sigma_*^2}{\mu\varepsilon}, \frac{\sigma_*}{\mu} \sqrt{\frac{n\hat{L}}{\varepsilon}} \right\} \log \frac{\mu \text{dist}^2(\mathbf{x}_1, \mathbb{X}_*)}{\varepsilon}$, our bound is better both in the low-accuracy regime when $\log \frac{\mu \text{dist}^2(\mathbf{x}_1, \mathbb{X}_*)}{\varepsilon} \leq \frac{\hat{L}}{\mu} \Leftrightarrow \frac{\mu \text{dist}^2(\mathbf{x}_1, \mathbb{X}_*)}{\exp(\hat{L}/\mu)} \leq \varepsilon$ and asymptotically in the high-accuracy regime as $\varepsilon \rightarrow 0$. Taken together, these comparisons indicate that our Corollary 2 improves on the best known bound for **SGD** under the quadratic growth condition in different regimes.

4 Theoretical Analysis: Upper Bound for **RR**

In this section, we lay the groundwork for proving Theorem 1 and complete its proof at the end. The section is organized into four parts. First, we provide the three most important lemmas in Subsection 4.1. Next, in Subsection 4.2, we establish an upper bound in Theorem 5, which indicates that **Shuffling SGD** under **RR** never converges more slowly than **SGD** under any reasonable stepsize. Then, Theorem 6 in Subsection 4.3 presents an alternative convergence rate for **Shuffling SGD** under **RR**, which demonstrates that Algorithm 1 under **RR** provably converges faster than **SGD** when the stepsize is sufficiently small. Finally, in Subsection 4.4, we conclude Theorem 1 and then use it to prove Corollary 1.

4.1 Three Core Lemmas

This subsection contains three core lemmas, all of which are critical to our analysis.

Before presenting the lemmas, we introduce two notions. Given a permutation π of $[n]$ and two indices $i, j \in [n]$ satisfying $j \leq i$, we define the following new permutation

$$\pi(i, j) \triangleq (\pi^1, \dots, \pi^{j-1}, \pi^i, \pi^{j+1}, \dots, \pi^{i-1}, \pi^j, \pi^{i+1}, \dots, \pi^n). \quad (3)$$

In words, $\pi(i, j)$ is the permutation generated by exchanging the elements π^i and π^j in π . Equipped with the notion of $\pi(i, j)$, we introduce the following virtual sequence, for any given $k \in [K]$,

$$\mathbf{x}_k^{l+1}(i, j) \triangleq \mathbf{x}_k^l(i, j) - \eta_k \nabla f_{\pi_k^l(i, j)}(\mathbf{x}_k^l(i, j)), \forall l \in [n], \quad \text{where } \mathbf{x}_k^1(i, j) \triangleq \mathbf{x}_k^1. \quad (4)$$

This means that the sequence $\mathbf{x}_k^l(i, j), \forall l \in [n+1]$ denotes the trajectory of the k -th epoch starting from $\mathbf{x}_k^1$, produced by Algorithm 1, but under the permutation $\pi_k(i, j)$. This virtual iterate can be viewed as a coupled sequence of the real output and, to the best of our knowledge, was first introduced by Sherman et al. (2021). It plays a fundamental role in our proof, as will become clear.

With these two concepts in hand, we proceed to state the three core lemmas. The first is Lemma 2, which is based on Lemma 2 of Sherman et al. (2021). For completeness, we reproduce the proof of Lemma 2 in Appendix C.

Lemma 2. *Given an arbitrary finite-sum function $\ell(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n \ell_i(\mathbf{x})$, suppose **RR** is employed, then for any $k \in [K]$ and $i \in [n]$, **Shuffling SGD** (Algorithm 1) guarantees that*

$$\mathbb{E} \left[\ell(\mathbf{x}_k^i) - \ell_{\pi_k^i}(\mathbf{x}_k^i) \right] = \frac{1}{n} \sum_{j=1}^{i-1} \mathbb{E} \left[\ell_{\pi_k^i}(\mathbf{x}_k^i(i, j)) - \ell_{\pi_k^i}(\mathbf{x}_k^i) \right],$$

where $\mathbf{x}_k(i, j)$ is defined in (4).

In the analysis of shuffling gradient methods, a well-known major challenge is to properly bound the term $\mathbb{E} [f(\mathbf{x}_k^i) - f_{\pi_k^i}(\mathbf{x}_k^i)]$, unlike in SGD, which no longer equals 0 due to the nature of shuffling-based algorithms. Lemma 2 provides a possible approach by relating the term we want to control (in a slightly more general form, applicable to any finite-sum function ℓ) to another quantity involving the virtual sequence introduced earlier in (4).

For the convenience of the discussion, temporarily assume $\ell_i = f_i$ in Lemma 2. Then, under the smoothness assumption, one would expect the difference between $\ell_{\pi_k^i}(\mathbf{x}_k^i(i, j))$ and $\ell_{\pi_k^i}(\mathbf{x}_k^i)$ to be small whenever $\mathbf{x}_k^i(i, j)$ and $\mathbf{x}_k^i$ are close. This observation naturally leads us to another core Lemma 3 stated below.

Lemma 3. *Under Assumptions 2 and 3, suppose $\eta_k \leq \frac{2}{L}, \forall k \in [K]$, then for any $k \in [K]$, $i \in [n]$, and $j \in [i - 1]$, Shuffling SGD (Algorithm 1) guarantees that*

$$\|\mathbf{x}_k^i(i, j) - \mathbf{x}_k^i\| \leq \eta_k \left\| \nabla f_{\pi_k^i}(\mathbf{x}_k^j) - \nabla f_{\pi_k^j}(\mathbf{x}_k^j) \right\|,$$

where $\mathbf{x}_k(i, j)$ is defined in (4).

Lemma 3 quantifies how close the virtual iterate and the true trajectory can be, under the widely required condition of $\eta_k \leq 2/\hat{L}$ in smooth optimization. Although the inequality does not directly offer a bound on the distance between $\mathbf{x}_k^i(i, j)$ and $\mathbf{x}_k^i$ that depends only on deterministic terms (e.g., the stepsize η_k), it is sufficient for our proof when combined with a careful analysis.

The proof of Lemma 3 is given in Appendix C and relies on the co-coercivity property (i.e., Lemma 1), which is closely related to the nonexpansiveness of the update rule in gradient methods (Polyak, 1987; Nesterov et al., 2018).

Equipped with Lemmas 2 and 3, we are ready to present the final core result, Lemma 4.

Lemma 4. *Under Assumptions 1, 2, and 3, suppose RR is employed with $\left\{ \eta_k \leq \frac{2}{L} \right\}_{k=1}^K$, then for any $k \in [K]$ and $i \in [n]$, Shuffling SGD (Algorithm 1) guarantees that*

$$\mathbb{E} [B(\mathbf{x}_k^i, \mathbf{x}_\star)] \leq 2\mathbb{E} [B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star)] + \frac{64}{n} \sum_{j=1}^{i-1} \mathbb{E} [B_{\pi_k^j}(\mathbf{x}_k^j, \mathbf{x}_\star)] + \frac{12(i-1)}{n} \eta_k^2 \hat{L} \sigma_\star^2.$$

Remark 6. We make no effort to optimize the constants in the bounds obtained in this work.

To the best of our knowledge, we are the first to obtain an inequality in such a form. Intuitively, Lemma 4 says that each $\mathbb{E} [B(\mathbf{x}_k^i, \mathbf{x}_\star)]$ (which now equals $\mathbb{E} [f(\mathbf{x}_k^i) - f_\star]$ because $\nabla f(\mathbf{x}_\star) = \mathbf{0}$) differs from its stochastic counterpart $\mathbb{E} [B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star)]$ by at most a multiplicative constant, together with an average of the preceding terms $\mathbb{E} [B_{\pi_k^j}(\mathbf{x}_k^j, \mathbf{x}_\star)]$ satisfying $j < i$ (not an exact average due to the coefficient $64/n$), plus an additional term involving $\sigma_\star^2$. In particular, summing the inequality in Lemma 4 from $i = 1$ to n yields a meaningful lower bound on $\mathbb{E} [\sum_{i=1}^n B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star)]$ based on $\mathbb{E} [\sum_{i=1}^n B(\mathbf{x}_k^i, \mathbf{x}_\star)]$ and $\sigma_\star^2$.

The proof of Lemma 4 is rather technical, so we skip the discussion here. For details, we kindly refer the interested reader to Appendix C.

4.2 Bound I: Never Worse than SGD under Reasonable Stepsize

In this subsection, we give the first bound for Shuffling SGD under RR, stated in Theorem 5 below.

Theorem 5. *Under Assumptions 1, 2, and 3, suppose RR is employed with $\left\{ \eta_k < \frac{2}{L} \right\}_{k=1}^K$, then Shuffling SGD (Algorithm 1) guarantees that*

$$\mathbb{E} [f(\bar{\mathbf{x}}_K) - f_\star] \lesssim \frac{\|\mathbf{x}_1^1 - \mathbf{x}_\star\|^2}{n \sum_{k=1}^K (2 - \eta_k \hat{L}) \eta_k} + \frac{\sum_{k=1}^K \eta_k^2 / (2 - \eta_k \hat{L})}{\sum_{k=1}^K (2 - \eta_k \hat{L}) \eta_k} \sigma_\star^2,$$

where $\bar{\mathbf{x}}_K$ is defined in (1).

Discussion on Theorem 5. To the best of our knowledge, Theorem 5 offers the first theoretical evidence that Shuffling SGD under RR shares surprising similarities with SGD, as reflected in the two aspects elaborated below.

First, Theorem 5 states that, similar to SGD, Shuffling SGD under RR does converge under any reasonable stepsize (i.e., $\eta_k < 2/\hat{L}$). In contrast, as far as we know, all prior works that provide provable rates of multi-epoch RR for smooth convex optimization require the stepsize η_k to be smaller than a threshold proportional to $1/n$, with only one exception (Nagaraj et al., 2019), which, however, assumes that each f_i is additionally Lipschitz, thereby limiting the applicability of their theory and even excluding common quadratic optimization problems over $\mathbb{R}^d$.

Second, we highlight that Theorem 5 gives the same convergence upper bound (up to constant factors) as SGD (Garrigos and Gower, 2023) in smooth convex optimization, while allowing a stepsize that depends on the epoch number. This result thus fills a gap in the literature.

Putting these together, Theorem 5 indicates that, for smooth convex optimization, RR under any reasonable stepsize never converges more slowly than SGD.

Analysis. In the following, we present the analysis for Theorem 5 and finally prove it. The core idea underlying the proof is, as one might expect, to analyze Shuffling SGD in a manner analogous to SGD. In other words, we aim to quantify the progress made by Algorithm 1 at each iteration. Although this perspective is natural, it has been less explored in prior studies. The main reason is that, as discussed earlier, the permutation in Shuffling SGD causes the most important property of SGD, unbiasedness, to no longer hold. To overcome this barrier, we develop a novel analysis that avoids any additional assumptions, such as Lipschitz continuity considered in Nagaraj et al. (2019).

We start with the following Lemma 5, which characterizes the per-iterate progress of Shuffling SGD (or SGD). Lemma 5 is inspired by Theorem 8 of Attia et al. (2025), which was established only for SGD. In contrast, our result slightly generalizes and tightens their bound. Indeed, if π_k^i is sampled uniformly from $[n]$, after taking expectations on both sides and dropping the nonnegative term $B_{\pi_k^i}(\mathbf{x}_\star, \mathbf{x}_k^i)$, Lemma 5 recovers their result. We include the proof of Lemma 5 in Appendix D.

Lemma 5. *Under Assumptions 1, 2, and 3, suppose $\eta_k < \frac{2}{\hat{L}}, \forall k \in [K]$, then for any $k \in [K]$ and $i \in [n]$, Shuffling SGD (Algorithm 1) guarantees that*

$$\begin{aligned} \frac{2 - \eta_k \hat{L}}{2} \eta_k B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) &\leq \|\mathbf{x}_k^i - \mathbf{x}_\star\|^2 - \|\mathbf{x}_k^{i+1} - \mathbf{x}_\star\|^2 + \frac{2 + \eta_k \hat{L}}{2 - \eta_k \hat{L}} \eta_k^2 \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 \\ &\quad - 2\eta_k \left\langle \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \right\rangle - \frac{2 - \eta_k \hat{L}}{2} \eta_k B_{\pi_k^i}(\mathbf{x}_\star, \mathbf{x}_k^i). \end{aligned}$$

In SGD, the uniform sampling scheme ensures that $\mathbb{E} [B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star)] = \mathbb{E} [B(\mathbf{x}_k^i, \mathbf{x}_\star)] = \mathbb{E} [f(\mathbf{x}_k^i) - f_\star]$ and $\mathbb{E} [\langle \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \rangle] = \mathbb{E} [\langle \nabla f(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \rangle] = 0$. In contrast, the analysis of Shuffling SGD presents two challenges. The first is to relate $\mathbb{E} [B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star)]$ to $\mathbb{E} [B(\mathbf{x}_k^i, \mathbf{x}_\star)]$, which can be handled by the previous Lemma 4. The second, and the main remaining difficulty, is to derive a suitable upper bound on $\mathbb{E} [-\langle \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \rangle]$.

We address this latter challenge by establishing the following novel inequality in Lemma 6.

Lemma 6. *Under Assumptions 1, 2, and 3, suppose RR is employed with $\left\{ \eta_k < \frac{2}{\hat{L}} \right\}_{k=1}^K$, then for any $k \in [K]$ and $i \in [n]$, Shuffling SGD (Algorithm 1) guarantees that*

$$\mathbb{E} [-\langle \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \rangle] \leq \frac{2 - \eta_k \hat{L}}{4n} \sum_{j=1}^{i-1} \mathbb{E} [B_{\pi_k^j}(\mathbf{x}_\star, \mathbf{x}_k^j)] + \frac{2(i-1)}{n} \cdot \frac{2 + 3\eta_k \hat{L}}{2 - \eta_k \hat{L}} \eta_k \sigma_\star^2.$$

Lemma 6 provides a novel inequality to upper bound $\mathbb{E} [-\langle \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \rangle]$ by the second moment of the stochastic gradients up to time i . Note that the second term on the R.H.S. can be absorbed by the

R.H.S. of the inequality in Lemma 5. For the remaining term, the coefficient η_k/n is key to the final proof, which ensures that the accumulated error in one epoch is controlled by $\mathbb{E}[\eta_k \sum_{i=1}^n \|\nabla f_{\pi_k^i}(\mathbf{x}_k^i)\|^2]$.

The proof of Lemma 6 builds on the two core results, Lemmas 2 and 3, presented before. To save space, we defer it to Appendix D.

Final proof. With Lemmas 5 and 6 stated above, we are ready to prove Theorem 5.

Proof of Theorem 5. We sum the inequality in Lemma 5 from $i = 1$ to n and use $\mathbf{x}_{k+1}^1 = \mathbf{x}_k^{n+1}$ to obtain that

$$\begin{aligned} \frac{2 - \eta_k \hat{L}}{2} \eta_k \sum_{i=1}^n B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) &\leq \|\mathbf{x}_k^1 - \mathbf{x}_\star\|^2 - \|\mathbf{x}_{k+1}^1 - \mathbf{x}_\star\|^2 + \frac{2 + \eta_k \hat{L}}{2 - \eta_k \hat{L}} \eta_k^2 \sum_{i=1}^n \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 \\ &\quad - 2\eta_k \sum_{i=1}^n \left\langle \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \right\rangle - \frac{2 - \eta_k \hat{L}}{2} \eta_k \sum_{i=1}^n B_{\pi_k^i}(\mathbf{x}_\star, \mathbf{x}_k^i). \end{aligned}$$

Take expectations on both sides and note that $\mathbb{E} \left[\left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 \right] = \sigma_\star^2$ to yield that

$$\begin{aligned} \frac{2 - \eta_k \hat{L}}{2} \eta_k \mathbb{E} \left[\sum_{i=1}^n B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) \right] &\leq \mathbb{E} \left[\|\mathbf{x}_k^1 - \mathbf{x}_\star\|^2 \right] - \mathbb{E} \left[\|\mathbf{x}_{k+1}^1 - \mathbf{x}_\star\|^2 \right] + \frac{2 + \eta_k \hat{L}}{2 - \eta_k \hat{L}} \eta_k^2 n \sigma_\star^2 \\ &\quad + 2\eta_k \sum_{i=1}^n \mathbb{E} \left[- \left\langle \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \right\rangle \right] - \frac{2 - \eta_k \hat{L}}{2} \eta_k \sum_{i=1}^n \mathbb{E} \left[B_{\pi_k^i}(\mathbf{x}_\star, \mathbf{x}_k^i) \right]. \end{aligned} \quad (5)$$

Next, we invoke Lemma 6 and sum it up from $i = 1$ to n to have

$$\sum_{i=1}^n \mathbb{E} \left[- \left\langle \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \right\rangle \right] \leq \frac{2 - \eta_k \hat{L}}{4} \sum_{i=1}^n \mathbb{E} \left[B_{\pi_k^i}(\mathbf{x}_\star, \mathbf{x}_k^i) \right] + \frac{2 + 3\eta_k \hat{L}}{2 - \eta_k \hat{L}} \eta_k n \sigma_\star^2. \quad (6)$$

Combine (5) and (6) to obtain that

$$\frac{2 - \eta_k \hat{L}}{2} \eta_k \mathbb{E} \left[\sum_{i=1}^n B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) \right] \leq \mathbb{E} \left[\|\mathbf{x}_k^1 - \mathbf{x}_\star\|^2 \right] - \mathbb{E} \left[\|\mathbf{x}_{k+1}^1 - \mathbf{x}_\star\|^2 \right] + \frac{6 + 7\eta_k \hat{L}}{2 - \eta_k \hat{L}} \eta_k^2 n \sigma_\star^2. \quad (7)$$

One more step, we observe that, by Lemma 4,

$$\mathbb{E} \left[\sum_{i=1}^n B(\mathbf{x}_k^i, \mathbf{x}_\star) \right] \leq 66 \mathbb{E} \left[\sum_{i=1}^n B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) \right] + 6\eta_k^2 n \hat{L} \sigma_\star^2 \leq 66 \mathbb{E} \left[\sum_{i=1}^n B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) \right] + 12\eta_k n \sigma_\star^2, \quad (8)$$

where the last step is due to $\eta_k \leq \frac{2}{\hat{L}}$.

Finally, we plug (8) back into (7), use $\eta_k \hat{L} \leq 2$, sum over $k = 1$ to K , divide both sides by $n \sum_{k=1}^K (2 - \eta_k \hat{L}) \eta_k$, apply the convexity of f , and use the definition of $\bar{\mathbf{x}}_K$ in (1) to conclude. $\square$

4.3 Bound II: Always Better than SGD under Small Stepsize

This subsection presents the other rate of Shuffling SGD under RR, as shown in Theorem 6 below.

Theorem 6. *Under Assumptions 1, 2, and 3, suppose RR is employed with $\left\{ \eta_k < \frac{2}{\hat{L}} \right\}_{k=1}^K$, then Shuffling SGD (Algorithm 1) guarantees that*

$$\mathbb{E} [f(\bar{\mathbf{x}}_K) - f_\star] \lesssim \frac{\|\mathbf{x}_1^1 - \mathbf{x}_\star\|^2}{n \sum_{k=1}^K (2 - \eta_k \hat{L}) \eta_k} + \frac{\sum_{k=1}^K \eta_k^3 n \bar{L} / (2 - \eta_k \hat{L})}{\sum_{k=1}^K (2 - \eta_k \hat{L}) \eta_k} \sigma_\star^2,$$

where $\bar{\mathbf{x}}_K$ is defined in (1).

Discussion on Theorem 6. Readers familiar with the literature on shuffling gradient methods may readily figure out that the rate given in Theorem 6 perfectly matches the known bound for **Shuffling SGD** under RR in smooth convex optimization (e.g., Mishchenko et al. (2020); Nguyen et al. (2021)). However, we emphasize a key difference here, that is, the stepsize in our Theorem 6 is allowed to satisfy $\eta_k < 2/\hat{L}$, in contrast to all existing results that require η_k to be at most inversely proportional to n .

More importantly, in the setting of nonuniform L_i considered in this work, the largest threshold on the stepsize in the literature that guarantees a rate similar to Theorem 6 is in the order of $1/(n\sqrt{\bar{L}\hat{L}})$ (Liu and Zhou, 2024; Cai et al., 2024). But as indicated by our Theorem 6, the superiority of RR over SGD already exists once $\eta_k \lesssim 1/(n\bar{L})$. Especially, this improvement can be significant when a dominant smoothness parameter exists, leading to $\hat{L} \approx n\bar{L}$.

Therefore, Theorem 6 is the first result to extend the known bound in smooth convex optimization to any reasonable stepsize while preserving the favorable property of Algorithm 1, i.e., **Shuffling SGD** under RR provably converges faster than SGD when the stepsize is sufficiently small.

Analysis. The roadmap for establishing Theorem 6 differs substantially from that used before to prove Theorem 5. This time, our proof strategy is to check how close **Shuffling SGD** can be to Gradient Descent. More concretely, we will view each epoch of Algorithm 1 (containing n iterations) as a single step and analyze the progress made by it at once. This kind of approach has appeared in different previous works (e.g., Mishchenko et al. (2020); Nguyen et al. (2021)) and always yields a convergence rate in the order of $1/(n^{1/3}K^{2/3})$, faster than SGD when K is large.

However, all works that follow the idea described above share the same issue, that is, they require the stepsize to be in the order of $1/n$, which is, however, not enough for our purpose. In the following, we develop a new analysis to bypass this critical obstacle.

We now formally begin the analysis by introducing another virtual sequence, defined as follows, for any $k \in [K]$,

$$\mathbf{y}_k^{i+1} \triangleq \mathbf{y}_k^i - \eta_k \nabla f_{\pi_k^i}(\mathbf{x}_*), \forall i \in [n], \quad \mathbf{y}_{k+1}^1 \triangleq \mathbf{y}_k^{n+1}, \quad \text{where } \mathbf{y}_1^1 \triangleq \mathbf{x}_*. \quad (9)$$

Under the above definition, and noting that $\nabla f(\mathbf{x}_*) = \mathbf{0}$, one can find

$$\mathbf{y}_k^{n+1} = \mathbf{y}_k^1 - \eta_k \sum_{i=1}^n \nabla f_{\pi_k^i}(\mathbf{x}_*) = \mathbf{y}_k^1 - \eta_k n \nabla f(\mathbf{x}_*) = \mathbf{y}_k^1, \forall k \in [K].$$

Combine the above line and $\mathbf{y}_1^1 = \mathbf{x}_*$ as defined in (9) to have

$$\mathbf{y}_k^{n+1} = \mathbf{y}_k^1 = \mathbf{x}_*, \forall k \in [K]. \quad (10)$$

The above virtual sequence is inspired by the work of Mishchenko et al. (2020), which, as far as we know, was the first to propose a similar term under the constant stepsize. Here, we slightly extend their idea to accommodate the case where the stepsize can depend on the current epoch number.

Remark 7. We note that Mishchenko et al. (2020) introduced the virtual sequence to handle the case of individual strong convexity, i.e., each f_i is required to be strongly convex. However, in our setting, only individual convexity is assumed. This means that their proof cannot be applied. As such, our analysis substantially departs from the existing approach.

Equipped with the new virtual sequence introduced above, we first present the following Lemma 7.

Lemma 7. *Under Assumptions 1, 2, and 3, suppose $\eta_k < \frac{2}{\hat{L}}, \forall k \in [K]$, then for any $k \in [K]$, **Shuffling SGD** (Algorithm 1) guarantees that*

$$\frac{4 - 2\eta_k \hat{L}}{2 + \eta_k \hat{L}} \eta_k \sum_{i=1}^n B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_*) \leq \|\mathbf{x}_k^1 - \mathbf{x}_*\|^2 - \|\mathbf{x}_{k+1}^1 - \mathbf{x}_*\|^2 + \frac{4 + \eta_k^2 \hat{L}^2}{4 - \eta_k^2 \hat{L}^2} \eta_k \sum_{i=1}^n L_{\pi_k^i} \|\mathbf{y}_k^i - \mathbf{x}_*\|^2,$$

where $\mathbf{y}_k^i$ is defined in (9).

As discussed earlier, we intentionally treat each epoch of **Shuffling SGD** as a single step. Hence, compared with Lemma 5 used to prove the first bound, Lemma 7 has a different flavor, which shows the progress made by Algorithm 1 over an entire epoch. Based on the form of Lemma 7, two tasks naturally arise. The first is to lower bound the term $\sum_{i=1}^n B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star)$ on the L.H.S. by $\sum_{i=1}^n B(\mathbf{x}_k^i, \mathbf{x}_\star) = \sum_{i=1}^n f(\mathbf{x}_k^i) - f_\star$. The second is to upper bound the residual term $\sum_{i=1}^n L_{\pi_k^i} \|\mathbf{y}_k^i - \mathbf{x}_\star\|^2$ on the R.H.S.

Fortunately, the first task can be directly addressed by summing the inequality in Lemma 4 over $i \in [n]$. Therefore, we only need to focus on the residual term $\sum_{i=1}^n L_{\pi_k^i} \|\mathbf{y}_k^i - \mathbf{x}_\star\|^2$ in Lemma 7, which is bounded in Lemma 8 below.

Lemma 8. *Under Assumptions 1 and 3, suppose RR is employed, then for any $k \in [K]$, **Shuffling SGD** (Algorithm 1) guarantees that*

$$\mathbb{E} \left[\sum_{i=1}^n L_{\pi_k^i} \|\mathbf{y}_k^i - \mathbf{x}_\star\|^2 \right] \leq \frac{\eta_k^2 n^2 \bar{L} \sigma_\star^2}{3},$$

where $\mathbf{y}_k^i$ is defined in (9).

Lemma 8 can be derived in a relatively easy way, as it can be deduced from existing works. In particular, thanks to Liu and Zhou (2024), we obtain a bound that depends only on the average smoothness parameter $\bar{L}$ rather than the maximum smoothness parameter $\hat{L}$.

Final proof. Armed with Lemmas 7 and 8 above, we are finally able to prove Theorem 6.

Proof of Theorem 6. First, we invoke Lemma 7 to have

$$\frac{4 - 2\eta_k \hat{L}}{2 + \eta_k \hat{L}} \eta_k \sum_{i=1}^n B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) \leq \|\mathbf{x}_k^1 - \mathbf{x}_\star\|^2 - \|\mathbf{x}_{k+1}^1 - \mathbf{x}_\star\|^2 + \frac{4 + \eta_k^2 \hat{L}^2}{4 - \eta_k^2 \hat{L}^2} \eta_k \sum_{i=1}^n L_{\pi_k^i} \|\mathbf{y}_k^i - \mathbf{x}_\star\|^2,$$

where $\mathbf{y}_k^i$ is defined in (9). Take expectations on both sides and apply Lemma 8 to yield

$$\frac{4 - 2\eta_k \hat{L}}{2 + \eta_k \hat{L}} \eta_k \mathbb{E} \left[\sum_{i=1}^n B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) \right] \leq \mathbb{E} [\|\mathbf{x}_k^1 - \mathbf{x}_\star\|^2] - \mathbb{E} [\|\mathbf{x}_{k+1}^1 - \mathbf{x}_\star\|^2] + \frac{4 + \eta_k^2 \hat{L}^2}{3(4 - \eta_k^2 \hat{L}^2)} \eta_k^3 n^2 \bar{L} \sigma_\star^2. \quad (11)$$

Next, we multiply both sides of the inequality in Lemma 4 by $\frac{4 - 2\eta_k \hat{L}}{2 + \eta_k \hat{L}} \eta_k$ and sum it up from $i = 1$ to n to obtain

$$\begin{aligned} \frac{4 - 2\eta_k \hat{L}}{2 + \eta_k \hat{L}} \eta_k \mathbb{E} \left[\sum_{i=1}^n B(\mathbf{x}_k^i, \mathbf{x}_\star) \right] &\leq 66 \frac{4 - 2\eta_k \hat{L}}{2 + \eta_k \hat{L}} \eta_k \mathbb{E} \left[\sum_{i=1}^n B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) \right] + \frac{24 - 12\eta_k \hat{L}}{2 + \eta_k \hat{L}} \eta_k^3 n \hat{L} \sigma_\star^2 \\ &\leq 66 \frac{4 - 2\eta_k \hat{L}}{2 + \eta_k \hat{L}} \eta_k \mathbb{E} \left[\sum_{i=1}^n B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) \right] + \frac{24 - 12\eta_k \hat{L}}{2 + \eta_k \hat{L}} \eta_k^3 n^2 \bar{L} \sigma_\star^2, \end{aligned} \quad (12)$$

where the last step is due to $\hat{L} \leq n\bar{L}$. Combine (11) and (12) to have

$$\frac{4 - 2\eta_k \hat{L}}{2 + \eta_k \hat{L}} \eta_k \mathbb{E} \left[\sum_{i=1}^n B(\mathbf{x}_k^i, \mathbf{x}_\star) \right] \leq 66 \mathbb{E} [\|\mathbf{x}_k^1 - \mathbf{x}_\star\|^2] - 66 \mathbb{E} [\|\mathbf{x}_{k+1}^1 - \mathbf{x}_\star\|^2] + \frac{12(2 - \eta_k \hat{L})^2 + 22(4 + \eta_k^2 \hat{L}^2)}{4 - \eta_k^2 \hat{L}^2} \eta_k^3 n^2 \bar{L} \sigma_\star^2,$$

which implies that, by $\eta_k \hat{L} \leq 2$,

$$\begin{aligned} \mathbb{E} \left[\sum_{i=1}^n (2 - \eta_k \hat{L}) \eta_k B(\mathbf{x}_k^i, \mathbf{x}_\star) \right] &\leq 132 \mathbb{E} [\|\mathbf{x}_k^1 - \mathbf{x}_\star\|^2] - 132 \mathbb{E} [\|\mathbf{x}_{k+1}^1 - \mathbf{x}_\star\|^2] + \frac{112 \eta_k^3 n^2 \bar{L}}{2 - \eta_k \hat{L}} \sigma_\star^2 \\ \Rightarrow \mathbb{E} \left[\sum_{k=1}^K \sum_{i=1}^n (2 - \eta_k \hat{L}) \eta_k B(\mathbf{x}_k^i, \mathbf{x}_\star) \right] &\leq 132 \|\mathbf{x}_1^1 - \mathbf{x}_\star\|^2 + 112 \sum_{k=1}^K \frac{\eta_k^3 n^2 \bar{L}}{2 - \eta_k \hat{L}} \sigma_\star^2. \end{aligned}$$

Finally, we observe that $B(\mathbf{x}_k^i, \mathbf{x}_\star) = f(\mathbf{x}_k^i) - f_\star$, divide both sides by $n \sum_{k=1}^K (2 - \eta_k \hat{L}) \eta_k$, apply the convexity of f , and use the definition of $\bar{\mathbf{x}}_K$ in (1) to conclude. $\square$

4.4 Proofs of Theorem 1 and Corollary 1

With the previous preparation, Theorem 1 follows immediately.

Proof of Theorem 1. Combine Theorems 5 and 6 to conclude. $\square$

We next derive Corollary 1 directly from Theorem 1.

Proof of Corollary 1. With a constant stepsize $\eta_k = \eta < \frac{2}{\bar{L}}, \forall k \in [K]$, Theorem 1 reduces to

$$\mathbb{E}[f(\bar{\mathbf{x}}_K) - f_\star] \lesssim \frac{\|\mathbf{x}_1^1 - \mathbf{x}_\star\|^2}{(2 - \eta\bar{L})\eta n K} + \frac{\min\{1, \eta n \bar{L}\} \eta \sigma_\star^2}{(2 - \eta\bar{L})^2}.$$

Optimizing the R.H.S. of the above inequality over $0 < \eta < \frac{2}{\bar{L}}$ yields the desired result. $\square$

5 Theoretical Analysis: Lower Bound for RR

In this section, our goal is to prove Theorem 2. Our strategy is again to separate the analysis into two complementary parts. In Subsection 5.1, we study the error arising from D , the initial distance between $\mathbf{x}_1^1$ and $\mathbf{x}_\star$, and derive the corresponding lower bound. In Subsection 5.2, we turn to the stochastic error induced by $\sigma_\star^2$, the variance of the gradient at $\mathbf{x}_\star$, and establish the second lower bound. Finally, in Subsection 5.3, we combine these two lower bounds to complete the proof of Theorem 2.

5.1 Bound I: Induced by the Initial Distance

In this subsection, we prove the first lower bound in Theorem 7, showing that the error induced by the initial distance D cannot decay faster than $\frac{D^2}{\eta n K}$ under the condition $K\eta n \bar{L} \geq 1$. Importantly, this lower bound applies to any shuffling strategy, rather than being specific to RR.

To prove the result, we construct a simple quadratic hard instance whose smoothness parameters for the individual components are chosen according to the prescribed stepsize η and parameters $L_1, \dots, L_n$. The resulting dynamics can be tracked explicitly, allowing us to analyze the iterates under an arbitrary ordering of the component functions and derive the claimed lower bound.

Theorem 7. *Given $d \in \mathbb{N}$, $n \geq 2$, $D > 0$, $\{L_i > 0\}_{i=1}^n$, and $\sigma_\star \geq 0$, suppose any shuffling strategy is employed with $\{\eta_k = \eta\}_{k=1}^K$ for $\eta > 0$, let $\bar{\mathbf{x}}_K \triangleq \frac{1}{nK} \sum_{k=1}^K \sum_{i=1}^n \mathbf{x}_k^i$ denote the average iterate generated by Shuffling SGD (Algorithm 1), initialized at $\mathbf{x}_1^1 = \mathbf{0}$, over $K \in \mathbb{N}$ epochs, then we have*

$$\sup_{f \in \mathcal{F}(d, n, D, L_{1:n}, \sigma_\star)} \mathbb{E}[f(\bar{\mathbf{x}}_K) - f_\star] \gtrsim \frac{D^2}{\eta n K} \quad \text{when } K\eta n \bar{L} \geq 1.$$

Remark 8. In fact, the lower bound is stronger than stated: for the constructed hard instance, the lower bound holds for every sample path, rather than merely in expectation. See the proof for details.

Proof. In the following proof, $\mathbf{v}$ denotes an arbitrary unit vector in $\mathbb{R}^d$. We first let

$$\beta_i \triangleq \min \left\{ \frac{1}{2\eta K}, n\bar{L} \right\} \frac{L_i}{n\bar{L}}, \forall i \in [n].$$

Clearly, each β_i satisfies $0 < \beta_i \leq \min \left\{ \frac{1}{\eta}, L_i \right\}$. Moreover, we write

$$\bar{\beta} \triangleq \frac{1}{n} \sum_{i=1}^n \beta_i = \min \left\{ \frac{1}{2\eta n K}, \bar{L} \right\}. \quad (13)$$

Next, we define

$$f_i(\mathbf{x}) \triangleq \frac{\beta_i \|\mathbf{x} - D\mathbf{v}\|^2}{2}, \forall i \in [n].$$

The corresponding finite-sum objective is

$$f(\mathbf{x}) \triangleq \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}) = \frac{\bar{\beta} \|\mathbf{x} - D\mathbf{v}\|^2}{2}. \quad (14)$$

Note that $\mathbf{x}_\star = D\mathbf{v} \Rightarrow \|\mathbf{x}_\star\| = D$, each f_i is differentiable, convex, and L_i -smooth (since $\beta_i \leq L_i$), and $\frac{1}{n} \sum_{i=1}^n \|\nabla f_i(\mathbf{x}_\star)\|^2 = 0 \leq \sigma_\star^2$. Therefore, f belongs to the class $\mathcal{F}(d, n, D, L_{1:n}, \sigma_\star)$.

Now, we can find that

$$\nabla f_{\pi_k^i}(\mathbf{x}_k^i) = \beta_{\pi_k^i}(\mathbf{x}_k^i - D\mathbf{v}), \forall k \in [K], i \in [n],$$

which implies that

$$\mathbf{x}_k^{i+1} = (1 - \eta\beta_{\pi_k^i})\mathbf{x}_k^i + \eta\beta_{\pi_k^i}D\mathbf{v}, \forall k \in [K], i \in [n].$$

Since $\mathbf{x}_1^1 = \mathbf{0}$, unrolling the recursion yields

$$\mathbf{x}_k^i = \left(1 - \rho^{k-1} \prod_{j=1}^{i-1} (1 - \eta\beta_{\pi_k^j})\right) D\mathbf{v}, \forall k \in [K], i \in [n],$$

where

$$\rho \triangleq \prod_{i=1}^n (1 - \eta\beta_i)^{\eta\beta_i \in [0,1], \forall i \in [n]} \geq 1 - \eta \sum_{i=1}^n \beta_i = 1 - \eta n \bar{\beta} \stackrel{(13)}{\geq} 1 - \frac{1}{2K} \geq \frac{1}{2}. \quad (15)$$

As such, we have

$$\bar{\mathbf{x}}_K = \frac{1}{nK} \sum_{k=1}^K \sum_{i=1}^n \mathbf{x}_k^i = \left(1 - \frac{1}{nK} \sum_{k=1}^K \rho^{k-1} \sum_{i=1}^n \prod_{j=1}^{i-1} (1 - \eta\beta_{\pi_k^j})\right) D\mathbf{v}. \quad (16)$$

Finally, we know

$$\begin{aligned} f(\bar{\mathbf{x}}_K) - f_\star &\stackrel{(14),(16), \|\mathbf{v}\|=1}{=} \frac{\bar{\beta} \left(\sum_{k=1}^K \rho^{k-1} \sum_{i=1}^n \prod_{j=1}^{i-1} (1 - \eta\beta_{\pi_k^j}) \right)^2 D^2}{2n^2 K^2} \stackrel{(a)}{\geq} \frac{(1 - \exp(-K\eta n \bar{\beta}))^2 D^2}{8\eta^2 n^2 K^2 \bar{\beta}} \\ &\stackrel{(b)}{\geq} \frac{\bar{\beta} D^2}{32} \stackrel{(13)}{=} \frac{D^2}{32\eta n K} \min \left\{ \frac{1}{2}, K\eta n \bar{L} \right\} \geq \frac{D^2}{64\eta n K} \min \{1, K\eta n \bar{L}\}, \end{aligned}$$

where (a) is due to

$$\prod_{j=1}^{i-1} (1 - \eta\beta_{\pi_k^j}) \geq \prod_{j=1}^n (1 - \eta\beta_j) \stackrel{(15)}{=} \rho, \forall i \in [n],$$

which implies that

$$\sum_{k=1}^K \rho^{k-1} \sum_{i=1}^n \prod_{j=1}^{i-1} (1 - \eta\beta_{\pi_k^j}) \geq n\rho \sum_{k=1}^K \rho^{k-1} \stackrel{(15)}{\geq} \frac{n}{2} \sum_{k=1}^K (1 - \eta n \bar{\beta})^{k-1} = \frac{1 - (1 - \eta n \bar{\beta})^K}{2\eta \bar{\beta}} \geq \frac{1 - \exp(-K\eta n \bar{\beta})}{2\eta \bar{\beta}},$$

and (b) follows from $1 - \exp(-x) \geq \frac{x}{2}, \forall x \in [0, \frac{1}{2}]$ and $0 \leq K\eta n \bar{\beta} \leq \frac{1}{2}$. Therefore, for the constructed $f \in \mathcal{F}(d, n, D, L_{1:n}, \sigma_\star)$, any shuffling strategy satisfies

$$f(\bar{\mathbf{x}}_K) - f_\star \gtrsim \frac{D^2}{\eta n K} \min \{1, K\eta n \bar{L}\}.$$

Consequently,

$$\sup_{f \in \mathcal{F}(d, n, D, L_{1:n}, \sigma_\star)} \mathbb{E}[f(\bar{\mathbf{x}}_K) - f_\star] \gtrsim \frac{D^2}{\eta n K} \quad \text{when } K\eta n \bar{L} \geq 1.$$

□

5.2 Bound II: Induced by the Gradient Variance at the Optimum

This subsection establishes the second lower bound in Theorem 8, which captures the stochastic error $\sigma_\star^2$ induced by the variance of the gradient at $\mathbf{x}_\star$. In contrast to the first lower bound in Theorem 7, which holds for any shuffling strategy, Theorem 8 is specific to RR and exploits the randomness of the permutation within each epoch.

Theorem 8. *Given $d \in \mathbb{N}$, $n \geq 2$, $D > 0$, $\{L_i > 0\}_{i=1}^n$, and $\sigma_\star \geq 0$, suppose RR is employed with $\{\eta_k = \eta\}_{k=1}^K$ for $\eta > 0$, let $\bar{\mathbf{x}}_K \triangleq \frac{1}{nK} \sum_{k=1}^K \sum_{i=1}^n \mathbf{x}_k^i$ denote the average iterate generated by Shuffling SGD (Algorithm 1), initialized at $\mathbf{x}_1^1 = \mathbf{0}$, over $K \in \mathbb{N}$ epochs, then we have*

$$\sup_{f \in \mathcal{F}(d, n, D, L_{1:n}, \sigma_\star)} \mathbb{E}[f(\bar{\mathbf{x}}_K) - f_\star] \gtrsim \min\{1, \eta n \bar{L}\} \eta \sigma_\star^2 \quad \text{when } K \eta n \bar{L} \geq 1.$$

The analysis required for Theorem 8 is considerably more involved. We therefore treat the cases $n \geq 3$ and $n = 2$ separately, establishing the corresponding results in Theorems 9 and 10 below, respectively.

At a high level, the proof is based on carefully designed hard instances whose dynamics under RR can be analyzed explicitly. For $n \geq 3$, a key ingredient is the asymmetric piecewise quadratic structure, allowing the randomness of the permutation to generate a persistent drift across epochs. Together with the nonuniform smoothness parameters of each component, this structure makes the effect of the random ordering analytically tractable. In contrast, the case $n = 2$ admits a simpler direct analysis. In both cases, the key is to quantify how the stochastic effect introduced within each epoch accumulates under RR and remains nonnegligible in the average iterate, ultimately yielding the claimed lower bound.

Analysis when $n \geq 3$. The key part of the proof is to design n asymmetric piecewise quadratic functions inspired by Rajput et al. (2020); Yun et al. (2022); Cha et al. (2023), in which sampling without replacement generates a persistent bias even when the algorithm starts at $\mathbf{x}_\star$. Controlling this bias across epochs then yields the desired term $\min\{1, \eta n \bar{L}\} \eta \sigma_\star^2$.

The most important technical component of our proof builds on Lemma 10 of Rajput et al. (2020), Lemma 13 of Yun et al. (2022), and Lemma B.4 of Cha et al. (2023). Concretely, given $k \in [K]$, we couple the true iterates $\{\mathbf{x}_k^i\}_{i=1}^{n+1}$ generated by the asymmetric piecewise quadratic hard instance $\{f_i\}_{i=1}^n$ (see (20)) with $\{\mathbf{z}_k^i\}_{i=1}^{n+1}$ generated by $\{h_i\}_{i=1}^n$ (see (22)), the symmetric counterpart of $\{f_i\}_{i=1}^n$, and then apply a novel comparison lemma (see Lemma 9) to derive a recursion in expectation for the true iterates. However, the existing analyses consider only uniform smoothness parameters, i.e., $L_i = L, \forall i \in [n]$. Therefore, our proof requires several modifications to accommodate the nonuniform case. Moreover, for our purpose, we must explicitly track the dependence on the stepsize η .

Theorem 9. *Given $d \in \mathbb{N}$, $n \geq 3$, $D > 0$, $\{L_i > 0\}_{i=1}^n$, and $\sigma_\star \geq 0$, suppose RR is employed with $\{\eta_k = \eta\}_{k=1}^K$ for $\eta > 0$, let $\bar{\mathbf{x}}_K \triangleq \frac{1}{nK} \sum_{k=1}^K \sum_{i=1}^n \mathbf{x}_k^i$ denote the average iterate generated by Shuffling SGD (Algorithm 1), initialized at $\mathbf{x}_1^1 = \mathbf{0}$, over $K \in \mathbb{N}$ epochs, then we have*

$$\sup_{f \in \mathcal{F}(d, n, D, L_{1:n}, \sigma_\star)} \mathbb{E}[f(\bar{\mathbf{x}}_K) - f_\star] \gtrsim \min\{1, \eta n \bar{L}\} \eta \sigma_\star^2 \quad \text{when } K \eta n \bar{L} \geq 1.$$

Proof. W.l.o.g., we assume $L_1 \geq L_2 \geq \dots \geq L_{n-1} \geq L_n$ in the following proof. We first define $S \triangleq \sum_{i=1}^{n-2} L_i$ and note that

$$3S \geq S + L_{n-1} + L_n = n\bar{L} \Rightarrow S \geq \frac{n\bar{L}}{3}. \quad (17)$$

We now introduce

$$\beta_i \triangleq \begin{cases} \min\left\{\frac{1}{2\eta}, S\right\} \frac{L_i}{S} & i \in [n-2] \\ 0 & i \in \{n-1, n\} \end{cases}. \quad (18)$$

By construction, each β_i satisfies $0 \leq \beta_i \leq \min\left\{\frac{1}{\eta}, L_i\right\}$ and at least one β_i is positive since $n \geq 3$. Moreover, we write

$$\bar{\beta} \triangleq \frac{1}{n} \sum_{i=1}^n \beta_i = \min\left\{\frac{1}{2\eta n}, \frac{S}{n}\right\} \stackrel{(17)}{\geq} \min\left\{\frac{1}{2\eta n}, \frac{\bar{L}}{3}\right\} \geq \frac{1}{3} \min\left\{\frac{1}{\eta n}, \bar{L}\right\}. \quad (19)$$

We then define

$$f_i(\mathbf{x}) \triangleq \left(\beta_i \mathbb{1}[\mathbf{x}[1] < 0] + \frac{\beta_i}{2} \mathbb{1}[\mathbf{x}[1] \geq 0] \right) \frac{(\mathbf{x}[1])^2}{2} + \chi_i \sqrt{\frac{n}{2}} \sigma_\star \mathbf{x}[1], \forall i \in [n], \quad (20)$$

where $\chi_i \triangleq \begin{cases} 0 & i \in [n-2] \\ -1 & i = n-1 \\ 1 & i = n \end{cases}$. The corresponding finite-sum objective is

$$f(\mathbf{x}) \triangleq \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}) = \left(\bar{\beta} \mathbb{1}[\mathbf{x}[1] < 0] + \frac{\bar{\beta}}{2} \mathbb{1}[\mathbf{x}[1] \geq 0] \right) \frac{(\mathbf{x}[1])^2}{2} \stackrel{(19)}{\geq} \frac{\min\left\{\frac{1}{\eta n}, \bar{L}\right\} (\mathbf{x}[1])^2}{12}. \quad (21)$$

Note that $\mathbf{x}_\star \triangleq \mathbf{0} \in \arg \min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x}) \Rightarrow \|\mathbf{x}_\star\| = 0 < D$, each f_i is differentiable, convex, and L_i -smooth since $\frac{\beta_1}{2} \leq \beta_i \leq L_i$, and $\frac{1}{n} \sum_{i=1}^n \|\nabla f_i(\mathbf{x}_\star)\|^2 = \frac{1}{2} \sum_{i=1}^n \chi_i^2 \sigma_\star^2 = \sigma_\star^2$. Therefore, $f \in \mathcal{F}(d, n, D, L_{1:n}, \sigma_\star)$.

Next, we introduce

$$h_i(\mathbf{x}) \triangleq \beta_i \frac{(\mathbf{x}[1])^2}{2} + \chi_i \sqrt{\frac{n}{2}} \sigma_\star \mathbf{x}[1], \forall i \in [n]. \quad (22)$$

Moreover, given $k \in [K]$, we recursively define

$$\mathbf{z}_k^{i+1} \triangleq \mathbf{z}_k^i - \eta \nabla h_{\pi_k^i}(\mathbf{z}_k^i), \forall i \in [n] \quad \text{where} \quad \mathbf{z}_k^1 \triangleq \mathbf{x}_k^1. \quad (23)$$

Furthermore, we let

$$\Delta_k^i \triangleq \mathbf{x}_k^i[1] - \mathbf{z}_k^i[1], \forall i \in [n].$$

We use induction to prove that, for all $i \in [n+1]$,

$$\Delta_k^i \geq \frac{\eta}{2} \sum_{j=1}^{i-1} \beta_{\pi_k^j} (\mathbf{z}_k^j[1])_+ \prod_{l=j+1}^{i-1} (1 - \eta \beta_{\pi_k^l}), \quad (24)$$

where $(x)_+ \triangleq x \mathbb{1}[x \geq 0]$. When $i = 1$, (24) is true by the definition of $\mathbf{z}_k^1 = \mathbf{x}_k^1$. Suppose (24) holds for some $i \in [n]$, then for the index $i+1$, we have

$$\begin{aligned} \mathbf{x}_k^{i+1}[1] &\stackrel{(20)}{=} \left(1 - \eta \beta_{\pi_k^i} \mathbb{1}[\mathbf{x}_k^i[1] < 0] - \eta \frac{\beta_{\pi_k^i}}{2} \mathbb{1}[\mathbf{x}_k^i[1] \geq 0] \right) \mathbf{x}_k^i[1] - \eta \chi_{\pi_k^i} \sqrt{\frac{n}{2}} \sigma_\star, \\ \mathbf{z}_k^{i+1}[1] &\stackrel{(22),(23)}{=} (1 - \eta \beta_{\pi_k^i}) \mathbf{z}_k^i[1] - \eta \chi_{\pi_k^i} \sqrt{\frac{n}{2}} \sigma_\star, \end{aligned}$$

which together imply that

$$\Delta_k^{i+1} = (1 - \eta \beta_{\pi_k^i}) \Delta_k^i + \frac{\eta}{2} \beta_{\pi_k^i} (\mathbf{x}_k^i[1])_+ \stackrel{(a)}{\geq} (1 - \eta \beta_{\pi_k^i}) \Delta_k^i + \frac{\eta}{2} \beta_{\pi_k^i} (\mathbf{z}_k^i[1])_+ \stackrel{(b)}{\geq} \frac{\eta}{2} \sum_{j=1}^i \beta_{\pi_k^j} (\mathbf{z}_k^j[1])_+ \prod_{l=j+1}^i (1 - \eta \beta_{\pi_k^l}),$$

where (a) is due to $\Delta_k^i \geq 0$ (by the inductive hypothesis) and the monotonicity of $(\cdot)_+$, and (b) is by $1 - \eta \beta_{\pi_k^i} \stackrel{(18)}{\geq} 1 - \frac{1}{2} > 0$ and the inductive hypothesis. Therefore, we complete the induction.

By (24), we have

$$\Delta_k^i \geq \frac{\eta}{2} \sum_{j=1}^{i-1} \beta_{\pi_k^j} (\mathbf{z}_k^j[1])_+ \prod_{l=j+1}^{i-1} (1 - \eta \beta_{\pi_k^l}) \geq \frac{\rho \eta}{2} \sum_{j=1}^{i-1} \beta_{\pi_k^j} (\mathbf{z}_k^j[1])_+, \forall i \in [n+1], \quad (25)$$

where

$$\rho \triangleq \prod_{i=1}^n (1 - \eta \beta_i) \stackrel{\eta \beta_i \in [0,1], \forall i \in [n]}{\geq} 1 - \eta \sum_{i=1}^n \beta_i = 1 - \eta n \bar{\beta} \stackrel{(19)}{=} 1 - \min\left\{\frac{1}{2}, \eta S\right\} \geq \frac{1}{2}. \quad (26)$$

Taking expectations on both sides of (25) yields that, for all $k \in [K]$ and $i \in [n+1]$,

$$\mathbb{E} [\mathbf{x}_k^i[1]] \geq \mathbb{E} [\mathbf{z}_k^i[1]] + \frac{\rho\eta}{2} \mathbb{E} \left[\sum_{j=1}^{i-1} \beta_{\pi_k^j} (\mathbf{z}_k^j[1])_+ \right] = \mathbb{E} \left[\prod_{j=1}^{i-1} (1 - \eta\beta_{\pi_k^j}) \right] \mathbb{E} [\mathbf{x}_k^1[1]] + \frac{\rho\eta}{2} \mathbb{E} \left[\sum_{j=1}^{i-1} \beta_{\pi_k^j} (\mathbf{z}_k^j[1])_+ \right], \quad (27)$$

where the last step is due to the first equation in Lemma 9.

Finally, we consider two cases:

Case 1. $K \geq 2$. In this case, we take $i = n+1$ in (27) to obtain, for all $k \in [K]$,

$$\begin{aligned} \mathbb{E} [\mathbf{x}_{k+1}^1[1]] &= \mathbb{E} [\mathbf{x}_k^{n+1}[1]] \geq \mathbb{E} \left[\prod_{j=1}^n (1 - \eta\beta_{\pi_k^j}) \right] \mathbb{E} [\mathbf{x}_k^1[1]] + \frac{\rho\eta}{2} \mathbb{E} \left[\sum_{i=1}^n \beta_{\pi_k^i} (\mathbf{z}_k^i[1])_+ \right] \\ &\stackrel{(26)}{=} \rho \mathbb{E} [\mathbf{x}_k^1[1]] + \frac{\rho\eta}{2} \mathbb{E} \left[\sum_{i=1}^n \beta_{\pi_k^i} (\mathbf{z}_k^i[1])_+ \right] \\ &\geq \rho \mathbb{E} [\mathbf{x}_k^1[1]] + \frac{\rho^2 \eta n \bar{\beta}}{12} \mathbb{E} \left[\left(\mathbf{x}_k^1[1] + \eta \sqrt{\frac{n}{2}} \sigma_\star \right)_+ \right], \end{aligned}$$

where the last step is by the second inequality in Lemma 9. Since $\mathbf{x}_1^1 = \mathbf{0}$, a simple induction gives that $\mathbb{E} [\mathbf{x}_k^1[1]] \geq 0, \forall k \in [K]$, which further implies that, for any $k \in [K]$,

$$\mathbb{E} [\mathbf{x}_{k+1}^1[1]] \geq \rho \mathbb{E} [\mathbf{x}_k^1[1]] + \frac{\rho^2 \eta^2 n^{\frac{3}{2}} \bar{\beta} \sigma_\star}{12\sqrt{2}} \stackrel{(26)}{\geq} (1 - \eta n \bar{\beta}) \mathbb{E} [\mathbf{x}_k^1[1]] + \frac{\rho^2 \eta^2 \bar{\beta} n^{\frac{3}{2}} \sigma_\star}{12\sqrt{2}}.$$

Unrolling this recursion yields

$$\mathbb{E} [\mathbf{x}_k^1[1]] \geq \frac{\rho^2 \eta \sqrt{n} \sigma_\star}{12\sqrt{2}} (1 - (1 - \eta n \bar{\beta})^{k-1}) \stackrel{(26)}{\geq} \frac{\eta \sqrt{n} \sigma_\star}{48\sqrt{2}} (1 - (1 - \eta n \bar{\beta})^{k-1}), \forall k \in [K]. \quad (28)$$

Next, we invoke (27) and drop the second nonnegative term to have

$$\mathbb{E} [\mathbf{x}_k^i[1]] \geq \mathbb{E} \left[\prod_{j=1}^{i-1} (1 - \eta\beta_{\pi_k^j}) \right] \mathbb{E} [\mathbf{x}_k^1[1]] \geq \frac{\mathbb{E} [\mathbf{x}_k^1[1]]}{2}, \forall k \in [K], i \in [n+1], \quad (29)$$

where the last step is due to $\mathbb{E} [\mathbf{x}_k^1[1]] \geq 0, \forall k \in [K]$ and $\prod_{j=1}^{i-1} (1 - \eta\beta_{\pi_k^j}) \stackrel{0 \leq \eta\beta_l \leq 1, \forall l \in [n]}{\geq} \prod_{j=1}^n (1 - \eta\beta_{\pi_k^j}) \stackrel{(26)}{\geq} \frac{1}{2}$. Therefore, we can find that

$$\begin{aligned} \mathbb{E} [\bar{\mathbf{x}}_K[1]] &= \frac{1}{nK} \sum_{k=1}^K \sum_{i=1}^n \mathbb{E} [\mathbf{x}_k^i[1]] \stackrel{(29)}{\geq} \frac{1}{2K} \sum_{k=1}^K \mathbb{E} [\mathbf{x}_k^1[1]] \stackrel{(28)}{\geq} \frac{\eta \sqrt{n} \sigma_\star}{96\sqrt{2}K} \sum_{k=1}^K 1 - (1 - \eta n \bar{\beta})^{k-1} \\ &\geq \frac{\eta \sqrt{n} \sigma_\star}{96\sqrt{2}K} \sum_{k=1}^K 1 - \exp(-\eta n \bar{\beta}(k-1)) \geq \frac{\eta \sqrt{n} \sigma_\star}{96\sqrt{2}K} \sum_{k=\lceil K/2 \rceil+1}^K 1 - \exp(-\eta n \bar{\beta}(k-1)) \\ &\geq \frac{\eta \sqrt{n} \sigma_\star}{96\sqrt{2}} \cdot \frac{\lfloor K/2 \rfloor}{K} \cdot (1 - \exp(-K\eta n \bar{\beta}/2)) \stackrel{K \geq 2}{\geq} \frac{\eta \sqrt{n} \sigma_\star}{288\sqrt{2}} (1 - \exp(-K\eta n \bar{\beta}/2)) \\ &\stackrel{(a)}{\geq} \frac{\min \left\{ 1 - \exp(-1/2), \min \left\{ \frac{K}{2}, \frac{K\eta n \bar{L}}{3} \right\} / 4 \right\}}{288\sqrt{2}} \eta \sqrt{n} \sigma_\star \geq \frac{\min \{1, K\eta n \bar{L}\}}{3456\sqrt{2}} \eta \sqrt{n} \sigma_\star, \quad (30) \end{aligned}$$

where (a) follows from $1 - \exp(-x) \geq \min\{1 - \exp(-1/2), x/2\}$, $\forall x \geq 0$ and $K\eta n\bar{\beta} \stackrel{(19)}{=} \min\left\{\frac{K}{2}, \frac{K\eta n\bar{L}}{3}\right\}$. Hence, we have

$$\begin{aligned}\mathbb{E}[f(\bar{\mathbf{x}}_K) - f_\star] &\geq f(\mathbb{E}[\bar{\mathbf{x}}_K]) - f_\star \stackrel{(21)}{\geq} \frac{\min\left\{\frac{1}{\eta n}, \bar{L}\right\} (\mathbb{E}[\bar{\mathbf{x}}_K[1]])^2}{12} \\ &\stackrel{(30)}{\gtrsim} \min\{1, \eta n\bar{L}\} (\min\{1, K\eta n\bar{L}\})^2 \eta \sigma_\star^2,\end{aligned}$$

where the first step is by Jensen's inequality.

Case 2. $K = 1$. In this case, we take $k = 1$ in (27) and use $\mathbf{x}_1^1 = \mathbf{0}$ to obtain

$$\mathbb{E}[\mathbf{x}_1^i[1]] \geq \frac{\rho\eta}{2} \mathbb{E}\left[\sum_{j=1}^{i-1} \beta_{\pi_1^j}(\mathbf{z}_1^j[1])_+\right], \forall i \in [n+1]. \quad (31)$$

Therefore, we can find that

$$\begin{aligned}\mathbb{E}[\bar{\mathbf{x}}_K[1]] &= \mathbb{E}[\bar{\mathbf{x}}_1[1]] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[\mathbf{x}_1^i[1]] \stackrel{(31)}{\geq} \frac{\rho\eta}{2n} \sum_{i=1}^n \mathbb{E}\left[\sum_{j=1}^{i-1} \beta_{\pi_1^j}(\mathbf{z}_1^j[1])_+\right] \\ &= \frac{\rho\eta}{2n} \mathbb{E}\left[\sum_{i=1}^n (n-i) \beta_{\pi_1^i}(\mathbf{z}_1^i[1])_+\right] \stackrel{(a)}{\geq} \frac{\rho^2 \eta (n-1) \bar{\beta}}{24} \mathbb{E}\left[\left(\mathbf{x}_1^1[1] + \eta \sqrt{\frac{n}{2}} \sigma_\star\right)_+\right] \\ &\stackrel{\mathbf{x}_1^1[1]=0}{=} \frac{\rho^2 \eta^2 (n-1) \sqrt{n} \bar{\beta} \sigma_\star}{24\sqrt{2}} \stackrel{(26)}{\geq} \frac{\eta^2 (n-1) \sqrt{n} \bar{\beta} \sigma_\star}{96\sqrt{2}} \stackrel{n-1 \geq \frac{n}{2}}{\geq} \frac{\eta^2 n^{\frac{3}{2}} \bar{\beta} \sigma_\star}{192\sqrt{2}} \\ &\stackrel{(26)}{\geq} \frac{\eta \sqrt{n} \min\{1, \eta n\bar{L}\} \sigma_\star}{576\sqrt{2}}.\end{aligned} \quad (32)$$

Hence, we have

$$\begin{aligned}\mathbb{E}[f(\bar{\mathbf{x}}_K) - f_\star] &\geq f(\mathbb{E}[\bar{\mathbf{x}}_K]) - f_\star \stackrel{(21)}{\geq} \frac{\min\left\{\frac{1}{\eta n}, \bar{L}\right\} (\mathbb{E}[\bar{\mathbf{x}}_K[1]])^2}{12} \\ &\stackrel{(32)}{\gtrsim} (\min\{1, \eta n\bar{L}\})^3 \eta \sigma_\star^2 \stackrel{K=1}{=} \min\{1, \eta n\bar{L}\} (\min\{1, K\eta n\bar{L}\})^2 \eta \sigma_\star^2,\end{aligned}$$

where the first step is by Jensen's inequality.

Combining the two cases above, we conclude that, for $n \geq 3$,

$$\sup_{f \in \mathcal{F}(d, n, D, L_{1:n}, \sigma_\star)} \mathbb{E}[f(\bar{\mathbf{x}}_K) - f_\star] \gtrsim \min\{1, \eta n\bar{L}\} (\min\{1, K\eta n\bar{L}\})^2 \eta \sigma_\star^2.$$

Consequently, for $n \geq 3$,

$$\sup_{f \in \mathcal{F}(d, n, D, L_{1:n}, \sigma_\star)} \mathbb{E}[f(\bar{\mathbf{x}}_K) - f_\star] \gtrsim \min\{1, \eta n\bar{L}\} \eta \sigma_\star^2 \quad \text{when} \quad K\eta n\bar{L} \geq 1.$$

□

Lemma 9 collects three technical results that characterize key properties of the auxiliary sequence $\{\mathbf{z}_k^i\}_{i=1}^{n+1}$ and serve as important ingredients for proving Theorem 9. In particular, these results provide the identities and lower bounds needed to control the drift induced by RR within each epoch.

Lemma 9. *Let $\{h_i\}_{i=1}^n$ be defined as in (22) and, for a fixed $k \in [K]$, let $\{\mathbf{z}_k^i\}_{i=1}^{n+1}$ be defined as in (23), then we have*

- $\mathbb{E} [\mathbf{z}_k^i[1]] = \mathbb{E} \left[\prod_{j=1}^{i-1} (1 - \eta \beta_{\pi_k^j}) \right] \mathbb{E} [\mathbf{x}_k^1[1]], \forall i \in [n+1].$
- $\mathbb{E} \left[\sum_{i=1}^n \beta_{\pi_k^i} (\mathbf{z}_k^i[1])_+ \right] \geq \frac{\rho n \bar{\beta}}{6} \mathbb{E} \left[(\mathbf{x}_k^1[1] + \eta \sqrt{\frac{n}{2}} \sigma_\star)_+ \right].$
- $\mathbb{E} \left[\sum_{i=1}^n (n-i) \beta_{\pi_k^i} (\mathbf{z}_k^i[1])_+ \right] \geq \frac{\rho n (n-1) \bar{\beta}}{12} \mathbb{E} \left[(\mathbf{x}_k^1[1] + \eta \sqrt{\frac{n}{2}} \sigma_\star)_+ \right].$

Proof. By definition,

$$\mathbf{z}_k^{i+1}[1] \stackrel{(22),(23)}{=} (1 - \eta \beta_{\pi_k^i}) \mathbf{z}_k^i[1] - \eta \chi_{\pi_k^i} \sqrt{\frac{n}{2}} \sigma_\star, \forall i \in [n].$$

Unrolling this recursion yields

$$\mathbf{z}_k^{i+1}[1] = \left(\prod_{j=1}^i (1 - \eta \beta_{\pi_k^j}) \right) \mathbf{x}_k^1[1] - \eta \sqrt{\frac{n}{2}} \sigma_\star \sum_{j=1}^i \chi_{\pi_k^j} \prod_{l=j+1}^i (1 - \eta \beta_{\pi_k^l}), \forall i \in [n]. \quad (33)$$

The first equation. For $\{a, b\} \subset [n]$, we define the operator $\mathcal{T}_{a \leftrightarrow b} : S_n \rightarrow S_n, o \mapsto \mathcal{T}_{a \leftrightarrow b}(o)$, where $\mathcal{T}_{a \leftrightarrow b}(o)$ is obtained by swapping the positions of a and b in $o \in S_n$. Recall that $\beta_{n-1} = \beta_n$ (see (18)) and

$$\chi_i = \begin{cases} 0 & i \in [n-2] \\ -1 & i = n-1 \\ 1 & i = n \end{cases}, \text{ we therefore have, for any } 1 \leq j \leq i \leq n,$$

$$\chi_{\mathcal{T}_{n-1 \leftrightarrow n}(\pi_k)} \prod_{l=j+1}^i (1 - \eta \beta_{\mathcal{T}_{n-1 \leftrightarrow n}(\pi_k)^l}) = \chi_{\mathcal{T}_{n-1 \leftrightarrow n}(\pi_k)} \prod_{l=j+1}^i (1 - \eta \beta_{\pi_k^l}) = -\chi_{\pi_k^j} \prod_{l=j+1}^i (1 - \eta \beta_{\pi_k^l}).$$

Consequently,

$$\mathbb{E} \left[\chi_{\pi_k^j} \prod_{l=j+1}^i (1 - \eta \beta_{\pi_k^l}) \right] = -\mathbb{E} \left[\chi_{\mathcal{T}_{n-1 \leftrightarrow n}(\pi_k)} \prod_{l=j+1}^i (1 - \eta \beta_{\mathcal{T}_{n-1 \leftrightarrow n}(\pi_k)^l}) \right] = -\mathbb{E} \left[\chi_{\pi_k^j} \prod_{l=j+1}^i (1 - \eta \beta_{\pi_k^l}) \right], \quad (34)$$

where the last step holds because, when π_k is uniformly sampled from S_n , π_k equals $\mathcal{T}_{n-1 \leftrightarrow n}(\pi_k)$ in distribution. Hence, (34) implies that

$$\mathbb{E} \left[\chi_{\pi_k^j} \prod_{l=j+1}^i (1 - \eta \beta_{\pi_k^l}) \right] = 0, \forall 1 \leq j \leq i \leq n.$$

Finally, we plug the above equation back into (33) to have

$$\mathbb{E} [\mathbf{z}_k^{i+1}[1]] = \mathbb{E} \left[\left(\prod_{j=1}^i (1 - \eta \beta_{\pi_k^j}) \right) \mathbf{x}_k^1[1] \right] = \mathbb{E} \left[\prod_{j=1}^i (1 - \eta \beta_{\pi_k^j}) \right] \mathbb{E} [\mathbf{x}_k^1[1]], \forall i \in [n],$$

where the last step follows from the independence of π_k and $\mathbf{x}_k^1$. Relabeling the indices completes the proof.

The second inequality. In what follows, for $i \in [n]$, let $\hat{\pi}_k^i \in [n]$ denote the unique index such that $\pi_k^{\hat{\pi}_k^i} = i$. In other words, $\hat{\pi}_k$ is the inverse permutation of π_k , and hence is also uniformly distributed over S_n . Moreover, we introduce the event $E_k^i \triangleq \{\hat{\pi}_k^{n-1} < \hat{\pi}_k^i < \hat{\pi}_k^n\}, \forall i \in [n-2]$. Since $\hat{\pi}_k$ is a random permutation, we have

$$\Pr [E_k^i] = \frac{1}{6}, \forall i \in [n-2]. \quad (35)$$

Now, we observe that

$$\sum_{i=1}^n \beta_{\pi_k^i} (\mathbf{z}_k^i[1])_+ = \sum_{i=1}^n \beta_i (\mathbf{z}_k^{\hat{\pi}_k^i}[1])_+ \stackrel{(a)}{=} \sum_{i=1}^{n-2} \beta_i (\mathbf{z}_k^{\hat{\pi}_k^i}[1])_+ \geq \sum_{i=1}^{n-2} \beta_i (\mathbf{z}_k^{\hat{\pi}_k^i}[1])_+ \mathbb{1} [E_k^i], \quad (36)$$

where (a) follows from $\beta_{n-1} = \beta_n = 0$ (see (18)). Next, given $i \in [n-2]$, we can find that

$$\begin{aligned}
(\mathbf{z}_k^{\hat{\pi}_k^i}[1])_+ \mathbb{1}[E_k^i] &\stackrel{(33)}{=} \left(\left[\prod_{j=1}^{\hat{\pi}_k^i-1} (1 - \eta\beta_{\pi_k^j}) \right] \mathbf{x}_k^1[1] - \eta\sqrt{\frac{n}{2}}\sigma_\star \sum_{j=1}^{\hat{\pi}_k^i-1} \chi_{\pi_k^j} \prod_{l=j+1}^{\hat{\pi}_k^i-1} (1 - \eta\beta_{\pi_k^l}) \right)_+ \mathbb{1}[E_k^i] \\
&\stackrel{(b)}{=} \left(\left[\prod_{j=1}^{\hat{\pi}_k^i-1} (1 - \eta\beta_{\pi_k^j}) \right] \mathbf{x}_k^1[1] + \eta\sqrt{\frac{n}{2}}\sigma_\star \prod_{l=\hat{\pi}_k^{n-1}+1}^{\hat{\pi}_k^i-1} (1 - \eta\beta_{\pi_k^l}) \right)_+ \mathbb{1}[E_k^i] \\
&\stackrel{(c)}{\geq} \left[\prod_{j=1}^n (1 - \eta\beta_{\pi_k^j}) \right] \left(\mathbf{x}_k^1[1] + \eta\sqrt{\frac{n}{2}}\sigma_\star \right)_+ \mathbb{1}[E_k^i] \stackrel{(26)}{=} \rho \left(\mathbf{x}_k^1[1] + \eta\sqrt{\frac{n}{2}}\sigma_\star \right)_+ \mathbb{1}[E_k^i], \quad (37)
\end{aligned}$$

where (b) holds because, under the event E_k^i , $\chi_{\pi_k^j} = 0$ for all $j \in \{1, \dots, \hat{\pi}_k^{n-1} - 1\} \cup \{\hat{\pi}_k^{n-1} + 1, \dots, \hat{\pi}_k^i - 1\}$ and $\chi_{\pi_k^j} = -1$ when $j = \hat{\pi}_k^{n-1}$, and (c) follows from combining the following two cases:

Case 1. $\mathbf{x}_k^1[1] \geq 0$. In this case, we have

$$\begin{aligned}
&\left(\left[\prod_{j=1}^{\hat{\pi}_k^i-1} (1 - \eta\beta_{\pi_k^j}) \right] \mathbf{x}_k^1[1] + \eta\sqrt{\frac{n}{2}}\sigma_\star \prod_{l=\hat{\pi}_k^{n-1}+1}^{\hat{\pi}_k^i-1} (1 - \eta\beta_{\pi_k^l}) \right)_+ \\
&= \left[\prod_{j=1}^{\hat{\pi}_k^i-1} (1 - \eta\beta_{\pi_k^j}) \right] \mathbf{x}_k^1[1] + \eta\sqrt{\frac{n}{2}}\sigma_\star \prod_{l=\hat{\pi}_k^{n-1}+1}^{\hat{\pi}_k^i-1} (1 - \eta\beta_{\pi_k^l}) \\
&\geq \left[\prod_{j=1}^n (1 - \eta\beta_{\pi_k^j}) \right] \left(\mathbf{x}_k^1[1] + \eta\sqrt{\frac{n}{2}}\sigma_\star \right) = \left[\prod_{j=1}^n (1 - \eta\beta_{\pi_k^j}) \right] \left(\mathbf{x}_k^1[1] + \eta\sqrt{\frac{n}{2}}\sigma_\star \right)_+.
\end{aligned}$$

Case 2. $\mathbf{x}_k^1[1] < 0$. In this case, we have

$$\begin{aligned}
&\left(\left[\prod_{j=1}^{\hat{\pi}_k^i-1} (1 - \eta\beta_{\pi_k^j}) \right] \mathbf{x}_k^1[1] + \eta\sqrt{\frac{n}{2}}\sigma_\star \prod_{l=\hat{\pi}_k^{n-1}+1}^{\hat{\pi}_k^i-1} (1 - \eta\beta_{\pi_k^l}) \right)_+ \\
&= \left[\prod_{j=\hat{\pi}_k^{n-1}+1}^{\hat{\pi}_k^i-1} (1 - \eta\beta_{\pi_k^j}) \right] \left(\left[\prod_{j=1}^{\hat{\pi}_k^{n-1}} (1 - \eta\beta_{\pi_k^j}) \right] \mathbf{x}_k^1[1] + \eta\sqrt{\frac{n}{2}}\sigma_\star \right)_+ \\
&\geq \left[\prod_{j=\hat{\pi}_k^{n-1}+1}^{\hat{\pi}_k^i-1} (1 - \eta\beta_{\pi_k^j}) \right] \left(\mathbf{x}_k^1[1] + \eta\sqrt{\frac{n}{2}}\sigma_\star \right)_+ \geq \left[\prod_{j=1}^n (1 - \eta\beta_{\pi_k^j}) \right] \left(\mathbf{x}_k^1[1] + \eta\sqrt{\frac{n}{2}}\sigma_\star \right)_+.
\end{aligned}$$

Finally, we plug (37) back into (36) to have

$$\sum_{i=1}^n \beta_{\pi_k^i} (\mathbf{z}_k^i[1])_+ \geq \rho \sum_{i=1}^{n-2} \beta_i \left(\mathbf{x}_k^1[1] + \eta\sqrt{\frac{n}{2}}\sigma_\star \right)_+ \mathbb{1}[E_k^i],$$

which implies that

$$\begin{aligned}
\mathbb{E} \left[\sum_{i=1}^n \beta_{\pi_k^i} (\mathbf{z}_k^i[1])_+ \right] &\geq \rho \sum_{i=1}^{n-2} \beta_i \mathbb{E} \left[\left(\mathbf{x}_k^1[1] + \eta\sqrt{\frac{n}{2}}\sigma_\star \right)_+ \mathbb{1}[E_k^i] \right] \\
&\stackrel{(d)}{=} \rho \sum_{i=1}^{n-2} \beta_i \Pr[E_k^i] \mathbb{E} \left[\left(\mathbf{x}_k^1[1] + \eta\sqrt{\frac{n}{2}}\sigma_\star \right)_+ \right] \\
&\stackrel{(19),(35)}{=} \frac{\rho n \bar{\beta}}{6} \mathbb{E} \left[\left(\mathbf{x}_k^1[1] + \eta\sqrt{\frac{n}{2}}\sigma_\star \right)_+ \right],
\end{aligned}$$

where (d) follows from the independence of π_k and $\mathbf{x}_k^1$.

The third inequality. Using an argument similar to that for the second inequality, we obtain

$$\begin{aligned} \mathbb{E} \left[\sum_{i=1}^n (n-i) \beta_{\pi_k^i} (\mathbf{z}_k^i[1])_+ \right] &\geq \rho \sum_{i=1}^{n-2} \beta_i \mathbb{E} [(n - \pi_k^i) \mathbb{1} [E_k^i]] \mathbb{E} \left[\left(\mathbf{x}_k^1[1] + \eta \sqrt{\frac{n}{2}} \sigma_\star \right)_+ \right] \\ &= \frac{\rho n(n-1)\bar{\beta}}{12} \mathbb{E} \left[\left(\mathbf{x}_k^1[1] + \eta \sqrt{\frac{n}{2}} \sigma_\star \right)_+ \right], \end{aligned}$$

where the last step is due to (19) and the fact that, for any $i \in [n-2]$,

$$\begin{aligned} \mathbb{E} [(n - \pi_k^i) \mathbb{1} [E_k^i]] &= n \Pr [E_k^i] - \mathbb{E} [\pi_k^i \mathbb{1} [E_k^i]] \stackrel{(35)}{=} \frac{n}{6} - \mathbb{E} [\pi_k^i \mathbb{1} [E_k^i]] \\ &= \frac{n}{6} - \sum_{l=2}^{n-1} \frac{l(l-1)(n-l)}{n(n-1)(n-2)} = \frac{n-1}{12}. \end{aligned}$$

□

Analysis when $n = 2$. We now turn to the case $n = 2$. Compared with the case $n \geq 3$, the construction and analysis are considerably simpler, as the problem involves only two component functions and the effect of the random permutation can be characterized explicitly. This allows us to establish the desired lower bound through a direct calculation.

Theorem 10. *Given $d \in \mathbb{N}$, $n = 2$, $D > 0$, $\{L_i > 0\}_{i=1}^n$, and $\sigma_\star \geq 0$, suppose RR is employed with $\{\eta_k = \eta\}_{k=1}^K$ for $\eta > 0$, let $\bar{\mathbf{x}}_K \triangleq \frac{1}{nK} \sum_{k=1}^K \sum_{i=1}^n \mathbf{x}_k^i$ denote the average iterate generated by **Shuffling SGD** (Algorithm 1), initialized at $\mathbf{x}_1^1 = \mathbf{0}$, over $K \in \mathbb{N}$ epochs, then we have*

$$\sup_{f \in \mathcal{F}(d, n, D, L_{1:n}, \sigma_\star)} \mathbb{E} [f(\bar{\mathbf{x}}_K) - f_\star] \gtrsim \min \{1, \eta n \bar{L}\} \eta \sigma_\star^2 \quad \text{when } K \eta n \bar{L} \geq 1.$$

Proof. W.l.o.g., we assume $L_1 \geq L_2$ in the following proof. We first define

$$\beta \triangleq \min \left\{ \frac{1}{\eta n}, \bar{L} \right\} = \min \left\{ \frac{1}{2\eta}, \bar{L} \right\}. \quad (38)$$

Next, we introduce

$$f_1(\mathbf{x}) \triangleq \beta \frac{(\mathbf{x}[1])^2}{2} - \sigma_\star \mathbf{x}[1] \quad \text{and} \quad f_2(\mathbf{x}) \triangleq \sigma_\star \mathbf{x}[1].$$

The corresponding finite-sum objective is

$$f(\mathbf{x}) \triangleq \frac{f_1(\mathbf{x}) + f_2(\mathbf{x})}{2} = \frac{\beta}{4} (\mathbf{x}[1])^2 \stackrel{(38)}{=} \frac{\min \left\{ \frac{1}{n\eta}, \bar{L} \right\} (\mathbf{x}[1])^2}{4}. \quad (39)$$

As before, one can verify that $f \in \mathcal{F}(d, n, D, L_{1:n}, \sigma_\star)$.

Again, we consider two cases:

Case 1. $K \geq 2$. In this case, a direct calculation gives us

$$\mathbb{E} [\mathbf{x}_{k+1}^1[1]] = (1 - \eta\beta) \mathbb{E} [\mathbf{x}_k^1[1]] + \frac{\eta^2 \beta \sigma_\star}{2}, \forall k \in [K].$$

Since $\mathbf{x}_1^1 = \mathbf{0}$, unrolling this recursion yields

$$\mathbb{E} [\mathbf{x}_k^1[1]] = \frac{\eta \sigma_\star}{2} (1 - (1 - \eta\beta)^{k-1}), \forall k \in [K+1]. \quad (40)$$

Next, we observe

$$\mathbb{E} [\mathbf{x}_k^2[1]] = \left(1 - \frac{\eta\beta}{2}\right) \mathbb{E} [\mathbf{x}_k^1[1]] \stackrel{\eta\beta \leq \frac{1}{2}}{\geq} \frac{3}{4} \mathbb{E} [\mathbf{x}_k^1[1]]. \quad (41)$$

Therefore, we can obtain that

$$\begin{aligned} \mathbb{E} [\bar{\mathbf{x}}_K[1]] &= \frac{1}{2K} \sum_{k=1}^K \sum_{i=1}^2 \mathbb{E} [\mathbf{x}_k^i[1]] \stackrel{(41)}{\geq} \frac{7}{8K} \sum_{k=1}^K \mathbb{E} [\mathbf{x}_k^1[1]] \stackrel{(40)}{\geq} \frac{7\eta\sigma_\star}{16K} \sum_{k=1}^K 1 - (1 - \eta\beta)^{k-1} \\ &\geq \frac{7\eta\sigma_\star}{16K} \sum_{k=1}^K 1 - \exp(-\eta\beta(k-1)) \geq \frac{7\eta\sigma_\star}{16K} \sum_{k=\lceil K/2 \rceil + 1}^K 1 - \exp(-\eta\beta(k-1)) \\ &\geq \frac{7\eta\sigma_\star}{16} \cdot \frac{\lfloor K/2 \rfloor}{K} \cdot (1 - \exp(-K\eta\beta/2)) \stackrel{K \geq 2}{\geq} \frac{7\eta\sigma_\star}{48} (1 - \exp(-K\eta\beta/2)) \\ &\stackrel{(a)}{\geq} \frac{7 \min \left\{ 1 - \exp(-1/2), \min \left\{ \frac{K}{2}, \frac{K\eta n \bar{L}}{2} \right\} / 4 \right\}}{48\sqrt{2}} \eta\sqrt{n}\sigma_\star \geq \frac{7 \min \{1, K\eta n \bar{L}\}}{384\sqrt{2}} \eta\sqrt{n}\sigma_\star, \end{aligned} \quad (42)$$

where (a) follows from $n = 2$, $1 - \exp(-x) \geq \min \{1 - \exp(-1/2), x/2\}$, $\forall x \geq 0$, and $K\eta\beta \stackrel{(38)}{=} \min \{ \frac{K}{2}, K\eta \bar{L} \} \stackrel{n=2}{=} \min \left\{ \frac{K}{2}, \frac{K\eta n \bar{L}}{2} \right\}$. Hence, we have

$$\begin{aligned} \mathbb{E} [f(\bar{\mathbf{x}}_K) - f_\star] &\geq f(\mathbb{E} [\bar{\mathbf{x}}_K]) - f_\star \stackrel{(39)}{\geq} \frac{\min \left\{ \frac{1}{n\eta}, \bar{L} \right\} (\mathbb{E} [\bar{\mathbf{x}}_K[1]])^2}{4} \\ &\stackrel{(42)}{\gtrsim} \min \{1, \eta n \bar{L}\} (\min \{1, K\eta n \bar{L}\})^2 \eta\sigma_\star^2, \end{aligned}$$

where the first step is by Jensen's inequality.

Case 2. $K = 1$. In this case, we know $\bar{\mathbf{x}}_K[1] = \bar{\mathbf{x}}_1[1] = \frac{\mathbf{x}_1^1[1] + \mathbf{x}_1^2[1]}{2}$ $\stackrel{\mathbf{x}_1^1[1]=0}{=} \frac{\mathbf{x}_1^2[1]}{2}$ and

$$\Pr [\mathbf{x}_1^2[1] = \eta\sigma_\star] = \Pr [\mathbf{x}_1^2[1] = -\eta\sigma_\star] = \frac{1}{2}.$$

Hence, we have

$$\begin{aligned} \mathbb{E} [f(\bar{\mathbf{x}}_K) - f_\star] &\stackrel{(39)}{\geq} \frac{\min \left\{ \frac{1}{\eta n}, \bar{L} \right\} \mathbb{E} [(\bar{\mathbf{x}}_K[1])^2]}{4} = \frac{\min \left\{ \frac{1}{\eta n}, \bar{L} \right\} \eta^2 \sigma_\star^2}{16} \\ &\stackrel{n=2}{=} \frac{\min \{1, \eta n \bar{L}\} \eta\sigma_\star^2}{32} \gtrsim \min \{1, \eta n \bar{L}\} \eta\sigma_\star^2 \\ &\geq \min \{1, \eta n \bar{L}\} (\min \{1, K\eta n \bar{L}\})^2 \eta\sigma_\star^2. \end{aligned}$$

Combining the two cases above, we conclude that, for $n = 2$,

$$\sup_{f \in \mathcal{F}(d, n, D, L_{1:n}, \sigma_\star)} \mathbb{E} [f(\bar{\mathbf{x}}_K) - f_\star] \gtrsim \min \{1, \eta n \bar{L}\} (\min \{1, K\eta n \bar{L}\})^2 \eta\sigma_\star^2.$$

Consequently, for $n = 2$,

$$\sup_{f \in \mathcal{F}(d, n, D, L_{1:n}, \sigma_\star)} \mathbb{E} [f(\bar{\mathbf{x}}_K) - f_\star] \gtrsim \min \{1, \eta n \bar{L}\} \eta\sigma_\star^2 \quad \text{when} \quad K\eta n \bar{L} \geq 1.$$

□

Final proof. With Theorems 9 and 10 in place, we can immediately prove Theorem 8.

Proof of Theorem 8. The result follows directly from Theorems 9 and 10. □

5.3 Proof of Theorem 2

The preceding results allow us to establish Theorem 2 now.

Proof of Theorem 2. Combining Theorems 7 and 8 completes the proof. $\square$

6 Theoretical Analysis: Lower Bound for SGD

In this section, we prove the algorithm-dependent lower bound for SGD with a fixed stepsize η , i.e., Theorem 3. Our proof proceeds by constructing a family of quadratic finite-sum problems tailored to η and showing that, for any such stepsize, SGD necessarily incurs an optimization error of the desired order. Similar to the previous proof for RR, the construction separates the effects of D (the initial distance to the minimizer) and $\sigma_\star^2$ (the variance of the gradient at the optimum), allowing us to derive the corresponding lower-bound terms individually and then combine them.

Compared with the analysis of the lower bound for RR, the case of SGD is relatively straightforward because the sampled indices are independent across iterations, making the calculations much easier. Nevertheless, the result is of independent interest: to the best of our knowledge, it provides the first algorithm-dependent lower bound for SGD that explicitly characterizes its performance for each prescribed stepsize η . In particular, rather than optimizing over the stepsize in advance, our lower bound holds for every fixed stepsize and thus directly captures the stepsize-dependent limitations of SGD.

Proof of Theorem 3. For simplicity, we write the update rule of SGD as

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta \nabla f_{\pi_t}(\mathbf{x}_t), \forall t \in [T] \quad \text{where} \quad T \triangleq nK,$$

where $\mathbf{x}_1 = \mathbf{0}$, $\eta > 0$, and π_t is sampled independently and uniformly from $[n]$. In the following proof, $\mathbf{v}$ denotes an arbitrary unit vector in $\mathbb{R}^d$.

We first define

$$\beta_i \triangleq \min \left\{ \frac{1}{2\eta T}, \bar{L} \right\} \frac{L_i}{\bar{L}}, \forall i \in [n].$$

Clearly, each β_i satisfies $0 < \beta_i \leq L_i$. Moreover, we write

$$\bar{\beta} \triangleq \frac{1}{n} \sum_{i=1}^n \beta_i = \min \left\{ \frac{1}{2\eta T}, \bar{L} \right\}. \quad (43)$$

We then fix $R \in [0, D]$ and $\nu \in [0, \sigma_\star]$, both of which will be decided later, and define

$$f_i(\mathbf{x}) \triangleq \frac{\beta_i \|\mathbf{x} - R\mathbf{v}\|^2}{2} + \chi_i \nu \langle \mathbf{v}, \mathbf{x} \rangle, \forall i \in [n],$$

where $\chi_i \triangleq \begin{cases} 1 & i \leq n \text{ is odd} \\ -1 & i \leq n \text{ is even} \end{cases}$ for even n , and $\chi_i \triangleq \begin{cases} 1 & i \leq n-1 \text{ is odd} \\ -1 & i \leq n-1 \text{ is even} \\ 0 & i = n \end{cases}$ for odd n . The corresponding

finite-sum objective is

$$f(\mathbf{x}) \triangleq \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}) = \frac{\bar{\beta} \|\mathbf{x} - R\mathbf{v}\|^2}{2}. \quad (44)$$

Note that $\mathbf{x}_\star = R\mathbf{v} \Rightarrow \|\mathbf{x}_\star\| = R \leq D$, each f_i is differentiable, convex, and L_i -smooth (since $\beta_i \leq L_i$), and $\frac{1}{n} \sum_{i=1}^n \|\nabla f_i(\mathbf{x}_\star)\|^2 = \frac{1}{n} \sum_{i=1}^n \chi_i^2 \nu^2 \leq \sigma_\star^2$. Therefore, f belongs to the class $\mathcal{F}(d, n, D, L_{1:n}, \sigma_\star)$.

We then have

$$\nabla f_{\pi_t}(\mathbf{x}_t) = \beta_{\pi_t}(\mathbf{x}_t - R\mathbf{v}) + \chi_{\pi_t} \nu \mathbf{v}, \forall t \in [T],$$

which implies that

$$\mathbf{x}_{t+1} = (1 - \eta\beta_{\pi_t})\mathbf{x}_t + \eta(\beta_{\pi_t}R - \chi_{\pi_t}\nu)\mathbf{v}, \forall t \in [T].$$

Since $\mathbf{x}_1 = \mathbf{0}$, unrolling the recursion yields that

$$\mathbf{x}_t = \left[\left(1 - \prod_{s=1}^{t-1} (1 - \eta\beta_{\pi_s}) \right) R - \sum_{s=1}^{t-1} \chi_{\pi_s} \prod_{r=s+1}^{t-1} (1 - \eta\beta_{\pi_r}) \eta\nu \right] \mathbf{v}, \forall t \in [T].$$

Therefore, we have

$$\bar{\mathbf{x}}_K = \frac{1}{T} \sum_{t=1}^T \mathbf{x}_t = \left[\left(1 - \frac{1}{T} \sum_{t=1}^T \prod_{s=1}^{t-1} (1 - \eta\beta_{\pi_s}) \right) R - \frac{1}{T} \sum_{t=1}^{T-1} \chi_{\pi_t} \sum_{s=t+1}^T \prod_{r=t+1}^{s-1} (1 - \eta\beta_{\pi_r}) \eta\nu \right] \mathbf{v}. \quad (45)$$

Next, we consider two cases:

Case 1. $R = D$ and $\nu = 0$. In this case, we have

$$\begin{aligned} \bar{\mathbf{x}}_K &\stackrel{(45)}{=} \left(1 - \frac{1}{T} \sum_{t=1}^T \prod_{s=1}^{t-1} (1 - \eta\beta_{\pi_s}) \right) D\mathbf{v} \\ \Rightarrow \mathbb{E}[\bar{\mathbf{x}}_K] &= \left(1 - \frac{1}{T} \sum_{t=1}^T \prod_{s=1}^{t-1} (1 - \eta\bar{\beta}) \right) D\mathbf{v} = \left(1 - \frac{1 - (1 - \eta\bar{\beta})^T}{\eta\bar{\beta}T} \right) D\mathbf{v}. \end{aligned} \quad (46)$$

Therefore, we know

$$\begin{aligned} \mathbb{E}[f(\bar{\mathbf{x}}_K) - f_\star] &\stackrel{(a)}{\geq} f(\mathbb{E}[\bar{\mathbf{x}}_K]) - f_\star \stackrel{(44), (46), \|\mathbf{v}\|=1}{=} \frac{(1 - (1 - \eta\bar{\beta})^T)^2 D^2}{2\eta^2 T^2 \bar{\beta}} \stackrel{\eta\bar{\beta} \in [0,1]}{\geq} \frac{(1 - \exp(-T\eta\bar{\beta}))^2 D^2}{2\eta^2 T^2 \bar{\beta}} \\ &\stackrel{(b)}{\geq} \frac{\bar{\beta} D^2}{8} \stackrel{(43)}{=} \frac{D^2}{8} \min \left\{ \frac{1}{2\eta T}, \bar{L} \right\} \geq \frac{D^2}{16\eta T} \min \{1, T\eta\bar{L}\} = \frac{D^2}{16\eta n K} \min \{1, K\eta n \bar{L}\}, \end{aligned}$$

where (a) is due to Jensen's inequality and (b) follows from $1 - \exp(-x) \geq \frac{x}{2}, \forall x \in [0, \frac{1}{2}]$ and $0 \leq T\eta\bar{\beta} \stackrel{(43)}{\leq} \frac{1}{2}$.

Case 2. $R = 0$ and $\nu = \sigma_\star$. In this case, we have

$$\bar{\mathbf{x}}_K \stackrel{(45)}{=} -\frac{1}{T} \sum_{t=1}^{T-1} \chi_{\pi_t} \sum_{s=t+1}^T \prod_{r=t+1}^{s-1} (1 - \eta\beta_{\pi_r}) \eta\sigma_\star \mathbf{v} = -\frac{1}{T} \sum_{t=1}^{T-1} \chi_{\pi_t} \alpha_t \eta\sigma_\star \mathbf{v}, \quad (47)$$

where $\alpha_t \triangleq \sum_{s=t+1}^T \prod_{r=t+1}^{s-1} (1 - \eta\beta_{\pi_r}), \forall t \in [T]$. Therefore, we know

$$\begin{aligned} \mathbb{E}[f(\bar{\mathbf{x}}_K) - f_\star] &\stackrel{(44), (47), \|\mathbf{v}\|=1}{=} \frac{\bar{\beta} \mathbb{E}[(\sum_{t=1}^{T-1} \chi_{\pi_t} \alpha_t)^2] \eta^2 \sigma_\star^2}{2T^2} \stackrel{(c)}{=} \frac{\bar{\beta} \sum_{t=1}^{T-1} \mathbb{E}[\chi_{\pi_t}^2 \alpha_t^2] \eta^2 \sigma_\star^2}{2T^2} \\ &\stackrel{(d)}{\geq} \frac{\sum_{t=1}^{T-1} (1 - \exp(-(T-t)\eta\bar{\beta}))^2 \sigma_\star^2}{3\bar{\beta}T^2} \stackrel{(e)}{\geq} \frac{(1 - \exp(-T\eta\bar{\beta}/2))^2 \sigma_\star^2}{9\bar{\beta}T} \stackrel{(f)}{\geq} \frac{\eta\sigma_\star^2}{144} \cdot T\eta\bar{\beta} \\ &\stackrel{(43)}{=} \frac{\eta\sigma_\star^2}{144} \min \left\{ \frac{1}{2}, T\eta\bar{L} \right\} \geq \frac{\eta\sigma_\star^2}{288} \min \{1, T\eta\bar{L}\} = \frac{\eta\sigma_\star^2}{288} \min \{1, K\eta n \bar{L}\}, \end{aligned}$$

where (c) is due to $\mathbb{E}[\chi_{\pi_t} \chi_{\pi_{t'}} \alpha_t \alpha_{t'}] = 0, \forall t \neq t' \in [T-1]$, (d) is by the independence of $\chi_{\pi_t}^2$ and α_t^2 , $\mathbb{E}[\chi_{\pi_t}^2] = \begin{cases} 1 & n \text{ is even} \\ \frac{n-1}{n} & n \text{ is odd} \end{cases} \geq \frac{2}{3}$ since $n \geq 2$, and

$$\mathbb{E}[\alpha_t^2] \geq (\mathbb{E}[\alpha_t])^2 = \left(\sum_{s=t+1}^T \prod_{r=t+1}^{s-1} (1 - \eta\bar{\beta}) \right)^2 = \frac{(1 - (1 - \eta\bar{\beta})^{T-t})^2}{\eta^2 \bar{\beta}^2} \stackrel{\eta\bar{\beta} \in [0,1]}{\geq} \frac{(1 - \exp(-(T-t)\eta\bar{\beta}))^2}{\eta^2 \bar{\beta}^2},$$

(e) holds by

$$\begin{aligned} \sum_{t=1}^{T-1} (1 - \exp(-(T-t)\eta\bar{\beta}))^2 &\geq \sum_{t=1}^{\lfloor \frac{T}{2} \rfloor} (1 - \exp(-(T-t)\eta\bar{\beta}))^2 \\ &\geq \left\lfloor \frac{T}{2} \right\rfloor (1 - \exp(-T\eta\bar{\beta}/2))^2 \stackrel{T=nK \geq 2}{\geq} \frac{T}{3} (1 - \exp(-T\eta\bar{\beta}/2))^2, \end{aligned}$$

and (f) follows from $1 - \exp(-x) \geq \frac{x}{2}, \forall x \in [0, \frac{1}{2}]$ and $0 \leq \frac{T\eta\bar{\beta}}{2} \stackrel{(43)}{\leq} \frac{1}{2}$.

Combining the two cases above, we conclude that

$$\sup_{f \in \mathcal{F}(d, n, D, L_{1:n}, \sigma_*)} \mathbb{E}[f(\bar{\mathbf{x}}_K) - f_*] \gtrsim \left(\frac{D^2}{\eta n K} + \eta \sigma_*^2 \right) \min \{1, K \eta n \bar{L}\}.$$

Consequently,

$$\sup_{f \in \mathcal{F}(d, n, D, L_{1:n}, \sigma_*)} \mathbb{E}[f(\bar{\mathbf{x}}_K) - f_*] \gtrsim \frac{D^2}{\eta n K} + \eta \sigma_*^2 \quad \text{when} \quad K \eta n \bar{L} \geq 1.$$

□

7 Theoretical Analysis: Application of RR

In this section, we first prove Theorem 4, which provides a convergence guarantee for Restarted Shuffling SGD with RR for functions satisfying the quadratic growth condition and serves as an application of our new Theorem 1. We then invoke Theorem 4 to establish Corollary 2, which gives the sample complexity, measured by the number of individual gradient evaluations, required to obtain an ε -accurate solution, i.e., to find a point $\mathbf{x} \in \mathbb{R}^d$ such that $\mathbb{E}[f(\mathbf{x}) - f_*] \lesssim \varepsilon$.

7.1 Proof of Theorem 4

Proof of Theorem 4. Note that f is real-valued. Hence, under Assumptions 1 and 2, $\mathbb{X}_* = \arg \min_{\mathbf{x} \in \mathbb{R}^d} f(\mathbf{x})$ is a nonempty closed convex set. Therefore, for any $\mathbf{x} \in \mathbb{R}^d$, the problem $\inf_{\mathbf{x}_* \in \mathbb{X}_*} \|\mathbf{x} - \mathbf{x}_*\|$ admits a unique solution, which we denote by $[\mathbf{x}]_* \triangleq \arg \min_{\mathbf{x}_* \in \mathbb{X}_*} \|\mathbf{x} - \mathbf{x}_*\|$ in the following. Moreover, we let $\mathcal{F}_t \triangleq \sigma(\text{permutations used in the first } t \text{ restart stages})$, i.e., the σ -algebra generated by the permutations in the first t restarts.

By Theorem 1, given $t \in [T]$ and any $\mathcal{F}_{t-1}$ -measurable $\mathbf{x}_* \in \mathbb{X}_*$,

$$\mathbb{E}[f(\mathbf{x}_{t+1}) - f_* \mid \mathcal{F}_{t-1}] \lesssim \frac{\|\mathbf{x}_t - \mathbf{x}_*\|^2}{(2 - \eta \hat{L}) \eta n K} + \frac{\min \{1, \eta n \bar{L}\} \eta \sigma_*^2}{(2 - \eta \hat{L})^2}.$$

Note that $\mathbf{x}_t$ is $\mathcal{F}_{t-1}$ -measurable, and hence $[\mathbf{x}_t]_*$ is also $\mathcal{F}_{t-1}$ -measurable. We can therefore replace $\mathbf{x}_*$ with $[\mathbf{x}_t]_*$ in the above inequality to obtain

$$\begin{aligned} \mathbb{E}[f(\mathbf{x}_{t+1}) - f_* \mid \mathcal{F}_{t-1}] &\lesssim \frac{\text{dist}^2(\mathbf{x}_t, \mathbb{X}_*)}{(2 - \eta \hat{L}) \eta n K} + \frac{\min \{1, \eta n \bar{L}\} \eta \sigma_*^2}{(2 - \eta \hat{L})^2} \\ &\Rightarrow \mathbb{E}[f(\mathbf{x}_{t+1}) - f_*] \lesssim \frac{\mathbb{E}[\text{dist}^2(\mathbf{x}_t, \mathbb{X}_*)]}{(2 - \eta \hat{L}) \eta n K} + \frac{\min \{1, \eta n \bar{L}\} \eta \sigma_*^2}{(2 - \eta \hat{L})^2}. \end{aligned}$$

Let $C > 0$ denote the hidden numerical constant in the above inequality (equivalently, the hidden constant in Theorem 1). When $K = \lceil 4CK_* \rceil = \left\lceil \frac{4C}{(2 - \eta \hat{L}) \eta n \mu} \right\rceil$, we have

$$\mathbb{E}[f(\mathbf{x}_{t+1}) - f_*] \leq \frac{\mu \mathbb{E}[\text{dist}^2(\mathbf{x}_t, \mathbb{X}_*)]}{4} + C \frac{\min \{1, \eta n \bar{L}\} \eta \sigma_*^2}{(2 - \eta \hat{L})^2}. \quad (48)$$

Next, by Assumption 4,

$$\frac{\mu}{2} \text{dist}^2(\mathbf{x}_{t+1}, \mathbb{X}_\star) \leq f(\mathbf{x}_{t+1}) - f_\star,$$

which implies that

$$\mathbb{E} [\text{dist}^2(\mathbf{x}_{t+1}, \mathbb{X}_\star)] \leq \frac{\mathbb{E} [\text{dist}^2(\mathbf{x}_t, \mathbb{X}_\star)]}{2} + 2C \frac{\min \{1, \eta n \bar{L}\} \eta \sigma_\star^2}{\mu(2 - \eta \hat{L})^2}.$$

Unrolling the above recursion gives

$$\mathbb{E} [\text{dist}^2(\mathbf{x}_t, \mathbb{X}_\star)] \leq \frac{\text{dist}^2(\mathbf{x}_1, \mathbb{X}_\star)}{2^{t-1}} + 4C \frac{\min \{1, \eta n \bar{L}\} \eta \sigma_\star^2}{\mu(2 - \eta \hat{L})^2}, \forall t \in [T+1]. \quad (49)$$

Finally, we plug (49) back into (48) to obtain that

$$\mathbb{E} [f(\mathbf{x}_{t+1}) - f_\star] \lesssim \frac{\mu \text{dist}^2(\mathbf{x}_1, \mathbb{X}_\star)}{2^t} + \frac{\min \{1, \eta n \bar{L}\} \eta \sigma_\star^2}{(2 - \eta \hat{L})^2}, \forall t \in [T].$$

In particular, taking $t = T$ in the above inequality yields the desired result. $\square$

7.2 Proof of Corollary 2

Proof of Corollary 2. By Theorem 4, we have

$$\mathbb{E} [f(\mathbf{x}_{T+1}) - f_\star] \lesssim \frac{\mu \text{dist}^2(\mathbf{x}_1, \mathbb{X}_\star)}{2^T} + \frac{\min \{1, \eta n \bar{L}\} \eta \sigma_\star^2}{(2 - \eta \hat{L})^2}.$$

When $T = T_\star = \left\lceil \log_2 \frac{\mu \text{dist}^2(\mathbf{x}_1, \mathbb{X}_\star)}{\varepsilon} \right\rceil$, we have

$$\frac{\mu \text{dist}^2(\mathbf{x}_1, \mathbb{X}_\star)}{2^{T_\star}} \leq \varepsilon.$$

When $\eta = \eta_\star = \min \left\{ \frac{1}{\bar{L}}, \max \left\{ \frac{\varepsilon}{\sigma_\star^2}, \frac{1}{\sigma_\star} \sqrt{\frac{\varepsilon}{n \bar{L}}} \right\} \right\}$, we observe

$$\frac{\min \{1, \eta_\star n \bar{L}\} \eta_\star \sigma_\star^2}{(2 - \eta_\star \hat{L})^2} \stackrel{\eta_\star \hat{L} \leq 1}{\leq} \min \{1, \eta_\star n \bar{L}\} \eta_\star \sigma_\star^2.$$

Next, we use $\eta_\star \leq \max \left\{ \frac{\varepsilon}{\sigma_\star^2}, \frac{1}{\sigma_\star} \sqrt{\frac{\varepsilon}{n \bar{L}}} \right\}$ to obtain that

$$\min \{1, \eta_\star n \bar{L}\} \leq \min \left\{ 1, \max \left\{ \frac{n \bar{L} \varepsilon}{\sigma_\star^2}, \frac{\sqrt{n \bar{L} \varepsilon}}{\sigma_\star} \right\} \right\} = \min \left\{ 1, \frac{\sqrt{n \bar{L} \varepsilon}}{\sigma_\star} \right\}$$

and

$$\eta_\star \sigma_\star^2 \leq \max \left\{ \varepsilon, \sigma_\star \sqrt{\frac{\varepsilon}{n \bar{L}}} \right\}.$$

The above three inequalities together imply that

$$\frac{\min \{1, \eta_\star n \bar{L}\} \eta_\star \sigma_\star^2}{(2 - \eta_\star \hat{L})^2} \leq \max \left\{ \varepsilon, \sigma_\star \sqrt{\frac{\varepsilon}{n \bar{L}}} \right\} \min \left\{ 1, \frac{\sqrt{n \bar{L} \varepsilon}}{\sigma_\star} \right\} \leq \varepsilon.$$

Therefore, we know

$$\mathbb{E} [f(\mathbf{x}_{T_\star+1}) - f_\star] \lesssim \varepsilon.$$

Finally, for sufficiently small ε such that $K_\star \geq 1$, the number of individual gradient evaluations is

$$\begin{aligned} nKT_\star &\asymp nK_\star T_\star \asymp n \cdot \frac{1}{(2 - \eta_\star \hat{L}) \eta_\star n \mu} \cdot \log \frac{\mu \text{dist}^2(\mathbf{x}_1, \mathbb{X}_\star)}{\varepsilon} \stackrel{\eta_\star \hat{L} \in [0,1]}{\asymp} \frac{1}{\eta_\star \mu} \log \frac{\mu \text{dist}^2(\mathbf{x}_1, \mathbb{X}_\star)}{\varepsilon} \\ &\asymp \left(\frac{\hat{L}}{\mu} + \min \left\{ \frac{\sigma_\star^2}{\mu \varepsilon}, \frac{\sigma_\star}{\mu} \sqrt{\frac{n \bar{L}}{\varepsilon}} \right\} \right) \log \frac{\mu \text{dist}^2(\mathbf{x}_1, \mathbb{X}_\star)}{\varepsilon}. \end{aligned}$$

$\square$

8 Conclusion and Future Work

In this work, we prove that Shuffling SGD under RR dominates standard SGD in smooth convex optimization under any reasonable stepsize after any finite number of epochs. Our Theorem 1 follows from combining two novel convergence results, whose analysis may each be of independent interest. We also establish new algorithm-dependent lower bounds for RR and SGD in Theorems 2 and 3. Together, these results provide (nearly) sharp worst-case characterizations of RR and standard SGD. Moreover, we combine our new convergence result with a restart framework and apply it to functions satisfying the quadratic growth condition, yielding a new convergence rate in Theorem 4 and a better sample complexity bound in Corollary 2 that improves upon the best known results, even those established under stronger assumptions.

Our work suggests several new directions for future research. First, our current proof of Theorem 1 is split into two distinct parts. It is therefore worthwhile to investigate whether Theorem 1 can be proved via a unified analysis. Second, the current upper and lower bounds for both RR and SGD differ by factors polynomial in $1/(2 - \eta\hat{L})$, and closing this gap remains an important direction. Finally, our results under the quadratic growth condition rely on a restart scheme. It would be interesting to determine whether Shuffling SGD can achieve comparable finite-time convergence guarantees without restarting.

Timeline and AI Disclosure

The improved stepsize threshold of $2/\hat{L}$ for the upper bounds in Theorem 1 and its application to quadratic-growth functions in Theorem 4 were obtained in March 2026 without any assistance from AI, shortly after the submission of the conference version (Liu, 2026), whose results were likewise proved without AI assistance. The lower bounds for RR and SGD in Theorems 2 and 3 were obtained in July 2026. The former was derived with assistance from GPT-5.6 Sol (particularly in establishing Theorem 8, a key intermediate result used in the proof of Theorem 2), whereas the latter was developed without AI assistance. GPT-5.6 Sol was also used to polish and refine the presentation during the final preparation of this extended version in September 2026.

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A Additional Related Work

This section provides additional discussion of the related work. We mainly focus on smooth convex optimization under RR. As for nonsmooth convex optimization under RR/SS/IG, the reader could refer to Kibardin (1979); Nedic and Bertsekas (2001); Koren et al. (2022); Liu and Zhou (2025). See also Shamir (2016) for RR under structured problems.

The first breakthrough on RR is by Gürbüzbalaban et al. (2021) for smooth strongly convex optimization, which, however, requires each f_i to be quadratic or to have a Lipschitz Hessian. Since then, extensive studies have emerged. Among them, a series of works continues to study the convergence behavior of RR in smooth strongly convex optimization for quadratic objectives or under other additional conditions (Ying et al., 2019; Haochen and Sra, 2019; Safran and Shamir, 2020; Rajput et al., 2020; Ahn et al., 2020; Safran and Shamir, 2021; Rajput et al., 2022).

To the best of our knowledge, the first work that drops the strong convexity assumption is Nagaraj et al. (2019), which provides a convergence rate of $\frac{D^2}{\eta n K} + \eta G^2$ under the requirement $\eta \leq \frac{2}{L}$ for L -smooth G -Lipschitz convex f_i , where we remind the reader that D denotes the distance between the initial point and the optimal solution, η represents the stepsize, and K is the number of epochs. However, this rate cannot reflect any advantage of RR over standard SGD. Subsequently, two works (Mishchenko et al., 2020; Nguyen et al., 2021) further remove the extra Lipschitz assumption and establish the bound $\frac{D^2}{\eta n K} + \eta^2 n L \sigma_\star^2$ under the condition $\eta \lesssim \frac{1}{nL}$, where $\sigma_\star^2$ is the gradient variance at the optimal solution. This rate is faster than standard SGD under its required regime and remains the best bound so far. In fact, it is unimprovable for small $\eta \lesssim \frac{1}{nL}$ and large $K \gtrsim \frac{nL^2 D^2}{\sigma_\star^2}$ due to the lower bound of $(\frac{L\sigma_\star^2 D^4}{nK^2})^{\frac{1}{3}}$ by Cha et al. (2023). Recently, Liu and Zhou (2024) and Cai and Diakonikolas (2025) extend the above rate from the average iterate to the last iterate (up to additional polylogarithmic factors).

B Summary of Notation

For readability, we recall and summarize the notation used in the paper.

- $f = \frac{1}{n} \sum_{i=1}^n f_i$ is the objective, where each $f_i : \mathbb{R}^d \rightarrow \mathbb{R}$ is differentiable and convex.
- $\mathbf{x}_\star$ denotes the minimizer of f . $f_\star = f(\mathbf{x}_\star)$ is the optimal function value. $\sigma_\star^2 = \frac{1}{n} \sum_{i=1}^n \|\nabla f_i(\mathbf{x}_\star)\|^2$ is the variance of the gradient at the optimal solution.
- $L_i > 0$ is the smoothness parameter of f_i . $\bar{L} = \frac{1}{n} \sum_{i=1}^n L_i$ is the average smoothness parameter. $\hat{L} = \max_{i \in [n]} L_i$ is the maximum smoothness parameter.
- $B_h(\mathbf{x}, \mathbf{y}) = h(\mathbf{x}) - h(\mathbf{y}) - \langle \nabla h(\mathbf{y}), \mathbf{x} - \mathbf{y} \rangle$ denotes the Bregman divergence induced by a real-valued differentiable function h (not necessarily convex). We write B (resp. B_i) as shorthand for B_f (resp. B_{f_i}).

C Missing Proofs of Core Lemmas

This section contains the missing proofs of the three core lemmas presented in Subsection 4.1.

Before providing the proofs, we recall two notions introduced in (3) and (4), respectively. Given a permutation π of $[n]$ and two indices $i, j \in [n]$ satisfying $j \leq i$, $\pi(i, j)$ is the permutation generated by exchanging the elements π_i and π_j in π , i.e.,

$$\pi(i, j) = (\pi^1, \dots, \pi^{j-1}, \pi^i, \pi^{j+1}, \dots, \pi^{i-1}, \pi^j, \pi^{i+1}, \dots, \pi^n). \quad (50)$$

For any given $k \in [K]$, $\mathbf{x}_k^l(i, j), \forall l \in [n+1]$ denotes the trajectory of the k -th epoch starting from $\mathbf{x}_k^1$, produced by Shuffling SGD, but under the permutation $\pi_k(i, j)$, i.e.,

$$\mathbf{x}_k^{l+1}(i, j) = \mathbf{x}_k^l(i, j) - \eta_k \nabla f_{\pi_k^l(i, j)}(\mathbf{x}_k^l(i, j)), \forall l \in [n], \quad \text{where } \mathbf{x}_k^1(i, j) = \mathbf{x}_k^1. \quad (51)$$

C.1 Proof of Lemma 2

Proof. Given $k \in [K]$ and $i \in [n]$, we first have the decomposition

$$\ell(\mathbf{x}_k^i) - \ell_{\pi_k^i}(\mathbf{x}_k^i) = \frac{1}{n} \left(\sum_{j=1}^{i-1} \ell_{\pi_k^j}(\mathbf{x}_k^i) - \ell_{\pi_k^i}(\mathbf{x}_k^i) + \sum_{j=i}^n \ell_{\pi_k^j}(\mathbf{x}_k^i) - \ell_{\pi_k^i}(\mathbf{x}_k^i) \right).$$

Note that $\mathbb{E}[\ell_{\pi_k^j}(\mathbf{x}_k^i)] = \mathbb{E}[\ell_{\pi_k^i}(\mathbf{x}_k^i)]$ holds for any $j \in \{i, \dots, n\}$ under RR, since $\ell_{\pi_k^j}(\mathbf{x}_k^i)$ and $\ell_{\pi_k^i}(\mathbf{x}_k^i)$ are equal in distribution conditioning on π_k^1 to π_k^{i-1} and π_1 to π_{k-1} . Therefore, we obtain

$$\mathbb{E}[\ell(\mathbf{x}_k^i) - \ell_{\pi_k^i}(\mathbf{x}_k^i)] = \frac{1}{n} \sum_{j=1}^{i-1} \mathbb{E}[\ell_{\pi_k^j}(\mathbf{x}_k^i) - \ell_{\pi_k^i}(\mathbf{x}_k^i)]. \quad (52)$$

For any fixed $j \in [i]$, it is known that π_k equals $\pi_k(i, j)$ in distribution (e.g., Lemma C.3 of Liu and Zhou (2025)), which implies that $(\pi_k, \mathbf{x}_k)$ also equals $(\pi_k(i, j), \mathbf{x}_k(i, j))$ in distribution, since the trajectory of the k -th epoch generated by Shuffling SGD is deterministically determined by the permutation and stepsize. This implies that

$$\mathbb{E}[\ell_{\pi_k^j}(\mathbf{x}_k^i)] = \mathbb{E}[\ell_{\pi_k^j(i, j)}(\mathbf{x}_k^i(i, j))] = \mathbb{E}[\ell_{\pi_k^i}(\mathbf{x}_k^i(i, j))],$$

where the last step is due to $\pi_k^j(i, j) = \pi_k^i$ from its definition (3). Hence, we finally obtain

$$\mathbb{E}[\ell(\mathbf{x}_k^i) - \ell_{\pi_k^i}(\mathbf{x}_k^i)] = \frac{1}{n} \sum_{j=1}^{i-1} \mathbb{E}[\ell_{\pi_k^i}(\mathbf{x}_k^i(i, j)) - \ell_{\pi_k^i}(\mathbf{x}_k^i)].$$

□

C.2 Proof of Lemma 3

Proof. By the definition of $\pi(i, j)$ in (3), we know

$$\pi_k^l(i, j) = \pi_k^l, \forall l \notin \{i, j\} \Rightarrow \nabla f_{\pi_k^l(i, j)} = \nabla f_{\pi_k^l}, \forall l \notin \{i, j\}. \quad (53)$$

Therefore, given $l \notin \{i, j\}$, by the definition of $\mathbf{x}_k(i, j)$ (see (4)) and the update rule of Shuffling SGD, we have

$$\begin{aligned} \|\mathbf{x}_k^{l+1}(i, j) - \mathbf{x}_k^{l+1}\|^2 &= \left\| \mathbf{x}_k^l(i, j) - \mathbf{x}_k^l - \eta_k \left(\nabla f_{\pi_k^l(i, j)}(\mathbf{x}_k^l(i, j)) - \nabla f_{\pi_k^l}(\mathbf{x}_k^l) \right) \right\|^2 \\ &\stackrel{(53)}{=} \left\| \mathbf{x}_k^l(i, j) - \mathbf{x}_k^l - \eta_k \left(\nabla f_{\pi_k^l}(\mathbf{x}_k^l(i, j)) - \nabla f_{\pi_k^l}(\mathbf{x}_k^l) \right) \right\|^2 \\ &= \left\| \mathbf{x}_k^l(i, j) - \mathbf{x}_k^l \right\|^2 + \eta_k^2 \left\| \nabla f_{\pi_k^l}(\mathbf{x}_k^l(i, j)) - \nabla f_{\pi_k^l}(\mathbf{x}_k^l) \right\|^2 \\ &\quad - 2\eta_k \left\langle \nabla f_{\pi_k^l}(\mathbf{x}_k^l(i, j)) - \nabla f_{\pi_k^l}(\mathbf{x}_k^l), \mathbf{x}_k^l(i, j) - \mathbf{x}_k^l \right\rangle \\ &\stackrel{(a)}{\leq} \left\| \mathbf{x}_k^l(i, j) - \mathbf{x}_k^l \right\|^2 + \left(\eta_k^2 - \frac{2\eta_k}{L_{\pi_k^l}} \right) \left\| \nabla f_{\pi_k^l}(\mathbf{x}_k^l(i, j)) - \nabla f_{\pi_k^l}(\mathbf{x}_k^l) \right\|^2 \\ &\stackrel{(b)}{\leq} \left\| \mathbf{x}_k^l(i, j) - \mathbf{x}_k^l \right\|^2, \end{aligned} \quad (54)$$

where (a) is due to Lemma 1 and (b) holds by $\eta_k \leq \frac{2}{L} \Rightarrow \eta_k^2 - \frac{2\eta_k}{L_{\pi_k^l}} \leq 0$.

Apply (54) from $l = j + 1$ to $l = i - 1$ to obtain

$$\begin{aligned} \left\| \mathbf{x}_k^i(i, j) - \mathbf{x}_k^i \right\|^2 &\leq \left\| \mathbf{x}_k^{j+1}(i, j) - \mathbf{x}_k^{j+1} \right\|^2 \\ &= \left\| \mathbf{x}_k^j(i, j) - \mathbf{x}_k^j - \eta_k \left(\nabla f_{\pi_k^j(i, j)}(\mathbf{x}_k^j(i, j)) - \nabla f_{\pi_k^j}(\mathbf{x}_k^j) \right) \right\|^2 \\ &\stackrel{(c)}{=} \eta_k^2 \left\| \nabla f_{\pi_k^j(i, j)}(\mathbf{x}_k^j) - \nabla f_{\pi_k^j}(\mathbf{x}_k^j) \right\|^2 \stackrel{(3)}{=} \eta_k^2 \left\| \nabla f_{\pi_k^i}(\mathbf{x}_k^j) - \nabla f_{\pi_k^j}(\mathbf{x}_k^j) \right\|^2, \end{aligned}$$

where (c) holds by $\mathbf{x}_k^j(i, j) = \mathbf{x}_k^j$, since $\mathbf{x}_k^l(i, j) = \mathbf{x}_k^l, \forall l \in [j]$ implied by (54) and $\mathbf{x}_k^1(i, j) \stackrel{(4)}{=} \mathbf{x}_k^1$ together. $\square$

C.3 Proof of Lemma 4

Proof. Expanding the definitions of B and $B_{\pi_k^i}$, we know

$$\begin{aligned} B(\mathbf{x}_k^i, \mathbf{x}_\star) - B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) &= f(\mathbf{x}_k^i) - f(\mathbf{x}_\star) - \langle \nabla f(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \rangle \\ &\quad - \left(f_{\pi_k^i}(\mathbf{x}_k^i) - f_{\pi_k^i}(\mathbf{x}_\star) - \langle \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \rangle \right) \\ &= f(\mathbf{x}_k^i) - f_{\pi_k^i}(\mathbf{x}_k^i) - f(\mathbf{x}_\star) + f_{\pi_k^i}(\mathbf{x}_\star) \\ &\quad - \langle \nabla f(\mathbf{x}_\star) - \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \rangle. \end{aligned}$$

Since $\mathbb{E}[f_{\pi_k^i}(\mathbf{x}_\star)] = f(\mathbf{x}_\star)$ and $\mathbb{E}[\nabla f_{\pi_k^i}(\mathbf{x}_\star)] = \nabla f(\mathbf{x}_\star)$ under RR, after taking expectations on both sides, we obtain that

$$\mathbb{E}[B(\mathbf{x}_k^i, \mathbf{x}_\star) - B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star)] = \mathbb{E}\left[f(\mathbf{x}_k^i) - f_{\pi_k^i}(\mathbf{x}_k^i) - \langle \nabla f(\mathbf{x}_\star) - \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i \rangle\right]. \quad (55)$$

Now, we denote by $\ell_i(\mathbf{x}) \triangleq f_i(\mathbf{x}) - \langle \nabla f_i(\mathbf{x}_\star), \mathbf{x} \rangle, \forall i \in [n]$ and $\ell(\mathbf{x}) \triangleq \frac{1}{n} \sum_{i=1}^n \ell_i(\mathbf{x}) = f(\mathbf{x}) - \langle \nabla f(\mathbf{x}_\star), \mathbf{x} \rangle$. Then, (55) implies that

$$\mathbb{E}[B(\mathbf{x}_k^i, \mathbf{x}_\star) - B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star)] = \mathbb{E}\left[\ell(\mathbf{x}_k^i) - \ell_{\pi_k^i}(\mathbf{x}_k^i)\right] \stackrel{(a)}{=} \frac{1}{n} \sum_{j=1}^{i-1} \mathbb{E}\left[\ell_{\pi_k^i}(\mathbf{x}_k^i(i, j)) - \ell_{\pi_k^i}(\mathbf{x}_k^i)\right], \quad (56)$$

where (a) holds by Lemma 2.

Note that $\ell_{\pi_k^i}$ is $L_{\pi_k^i}$ -smooth by its definition and Assumption 3, we therefore have, for any $j \in [i-1]$,

$$\begin{aligned} \ell_{\pi_k^i}(\mathbf{x}_k^i(i, j)) - \ell_{\pi_k^i}(\mathbf{x}_k^i) &\leq \langle \nabla \ell_{\pi_k^i}(\mathbf{x}_k^i), \mathbf{x}_k^i(i, j) - \mathbf{x}_k^i \rangle + \frac{L_{\pi_k^i}}{2} \|\mathbf{x}_k^i(i, j) - \mathbf{x}_k^i\|^2 \\ &= \langle \nabla f_{\pi_k^i}(\mathbf{x}_k^i) - \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i(i, j) - \mathbf{x}_k^i \rangle + \frac{L_{\pi_k^i}}{2} \|\mathbf{x}_k^i(i, j) - \mathbf{x}_k^i\|^2 \\ &\stackrel{(b)}{\leq} \frac{\left\| \nabla f_{\pi_k^i}(\mathbf{x}_k^i) - \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2}{2L_{\pi_k^i}} + L_{\pi_k^i} \|\mathbf{x}_k^i(i, j) - \mathbf{x}_k^i\|^2 \\ &\stackrel{(c)}{\leq} B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) + L_{\pi_k^i} \|\mathbf{x}_k^i(i, j) - \mathbf{x}_k^i\|^2 \\ &\stackrel{(d)}{\leq} B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) + \eta_k^2 L_{\pi_k^i} \left\| \nabla f_{\pi_k^i}(\mathbf{x}_k^j) - \nabla f_{\pi_k^j}(\mathbf{x}_k^j) \right\|^2, \end{aligned} \quad (57)$$

where (b) is by Cauchy-Schwarz inequality and AM-GM inequality, (c) is due to Lemma 1, and (d) holds by Lemma 3. Furthermore, we can bound

$$\begin{aligned} &\left\| \nabla f_{\pi_k^i}(\mathbf{x}_k^j) - \nabla f_{\pi_k^j}(\mathbf{x}_k^j) \right\|^2 \\ &\leq 4 \left\| \nabla f_{\pi_k^i}(\mathbf{x}_k^j) - \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 + 4 \left\| \nabla f_{\pi_k^j}(\mathbf{x}_k^j) - \nabla f_{\pi_k^j}(\mathbf{x}_\star) \right\|^2 + 4 \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 + 4 \left\| \nabla f_{\pi_k^j}(\mathbf{x}_\star) \right\|^2 \\ &\leq 8L_{\pi_k^i} B_{\pi_k^i}(\mathbf{x}_k^j, \mathbf{x}_\star) + 8L_{\pi_k^j} B_{\pi_k^j}(\mathbf{x}_k^j, \mathbf{x}_\star) + 4 \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 + 4 \left\| \nabla f_{\pi_k^j}(\mathbf{x}_\star) \right\|^2, \end{aligned}$$

where the last step is by Lemma 1 again. Plug the above inequality back into (57) and use $\eta_k \leq \frac{2}{L}$ to have

$$\begin{aligned} \ell_{\pi_k^i}(\mathbf{x}_k^i(i, j)) - \ell_{\pi_k^i}(\mathbf{x}_k^i) &\leq B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) + 32B_{\pi_k^i}(\mathbf{x}_k^j, \mathbf{x}_\star) + 32B_{\pi_k^j}(\mathbf{x}_k^j, \mathbf{x}_\star) \\ &\quad + 4\eta_k^2 L_{\pi_k^i} \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 + 4\eta_k^2 L_{\pi_k^i} \left\| \nabla f_{\pi_k^j}(\mathbf{x}_\star) \right\|^2. \end{aligned} \quad (58)$$

Combine (56) and (58) to obtain

$$\begin{aligned} \mathbb{E} \left[B(\mathbf{x}_k^i, \mathbf{x}_\star) - B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) \right] &\leq \frac{1}{n} \sum_{j=1}^{i-1} \mathbb{E} \left[B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) + 32B_{\pi_k^i}(\mathbf{x}_k^j, \mathbf{x}_\star) + 32B_{\pi_k^j}(\mathbf{x}_k^j, \mathbf{x}_\star) \right] \\ &\quad + \frac{4\eta_k^2}{n} \sum_{j=1}^{i-1} \mathbb{E} \left[L_{\pi_k^i} \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 + L_{\pi_k^i} \left\| \nabla f_{\pi_k^j}(\mathbf{x}_\star) \right\|^2 \right]. \end{aligned}$$

When $j < i$, we observe that $\mathbb{E} \left[B_{\pi_k^i}(\mathbf{x}_k^j, \mathbf{x}_\star) \right] = \mathbb{E} \left[B_{\pi_k^j}(\mathbf{x}_k^j, \mathbf{x}_\star) \right]$ and the following two inequalities hold

$$\begin{aligned} \mathbb{E} \left[L_{\pi_k^i} \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 \right] &= \frac{\sum_{l=1}^n L_l \left\| \nabla f_l(\mathbf{x}_\star) \right\|^2}{n} \leq \hat{L}\sigma_\star^2, \\ \mathbb{E} \left[L_{\pi_k^i} \left\| \nabla f_{\pi_k^j}(\mathbf{x}_\star) \right\|^2 \right] &= \frac{\sum_{l=1}^n \frac{n\bar{L}-L_l}{n-1} \left\| \nabla f_l(\mathbf{x}_\star) \right\|^2}{n} \leq 2\bar{L}\sigma_\star^2 \leq 2\hat{L}\sigma_\star^2. \end{aligned}$$

Put everything together, rearrange terms, and use $i-1 \leq n$ to conclude the desired inequality. $\square$

D Missing Proofs of Lemmas for Bound I

In this section, we provide the missing proofs of the lemmas presented in Subsection 4.2, which were used to prove the first convergence rate in Theorem 5.

D.1 Proof of Lemma 5

Proof. Given $k \in [K]$ and $i \in [n]$, by the update rule of Algorithm 1, we have

$$\begin{aligned} \left\| \mathbf{x}_k^{i+1} - \mathbf{x}_\star \right\|^2 &= \left\| \mathbf{x}_k^i - \mathbf{x}_\star - \eta_k \nabla f_{\pi_k^i}(\mathbf{x}_k^i) \right\|^2 \\ &= \left\| \mathbf{x}_k^i - \mathbf{x}_\star \right\|^2 + \eta_k^2 \left\| \nabla f_{\pi_k^i}(\mathbf{x}_k^i) \right\|^2 - 2\eta_k \left\langle \nabla f_{\pi_k^i}(\mathbf{x}_k^i), \mathbf{x}_k^i - \mathbf{x}_\star \right\rangle. \end{aligned} \quad (59)$$

Note that $\eta_k < \frac{2}{\bar{L}} \Rightarrow 2 - \eta_k L_{\pi_k^i} > 0$, we hence can bound

$$\begin{aligned} \left\| \nabla f_{\pi_k^i}(\mathbf{x}_k^i) \right\|^2 &\leq \left(1 + \frac{2 - \eta_k L_{\pi_k^i}}{2\eta_k L_{\pi_k^i}} \right) \left\| \nabla f_{\pi_k^i}(\mathbf{x}_k^i) - \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 + \left(1 + \frac{2\eta_k L_{\pi_k^i}}{2 - \eta_k L_{\pi_k^i}} \right) \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 \\ &= \frac{2 + \eta_k L_{\pi_k^i}}{2\eta_k L_{\pi_k^i}} \left\| \nabla f_{\pi_k^i}(\mathbf{x}_k^i) - \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 + \frac{2 + \eta_k L_{\pi_k^i}}{2 - \eta_k L_{\pi_k^i}} \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 \\ &\leq \frac{2 + \eta_k L_{\pi_k^i}}{2\eta_k} \left\langle \nabla f_{\pi_k^i}(\mathbf{x}_k^i) - \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \right\rangle + \frac{2 + \eta_k L_{\pi_k^i}}{2 - \eta_k L_{\pi_k^i}} \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2, \end{aligned} \quad (60)$$

where the last step is due to Lemma 1.

We plug (60) back into (59) to obtain that

$$\begin{aligned} \left\| \mathbf{x}_k^{i+1} - \mathbf{x}_\star \right\|^2 &\leq \left\| \mathbf{x}_k^i - \mathbf{x}_\star \right\|^2 - \frac{2 - \eta_k L_{\pi_k^i}}{2} \eta_k \left\langle \nabla f_{\pi_k^i}(\mathbf{x}_k^i) - \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \right\rangle \\ &\quad - 2\eta_k \left\langle \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \right\rangle + \frac{2 + \eta_k L_{\pi_k^i}}{2 - \eta_k L_{\pi_k^i}} \eta_k^2 \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 \\ &= \left\| \mathbf{x}_k^i - \mathbf{x}_\star \right\|^2 - \frac{2 - \eta_k L_{\pi_k^i}}{2} \eta_k \left(B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) + B_{\pi_k^i}(\mathbf{x}_\star, \mathbf{x}_k^i) \right) \\ &\quad - 2\eta_k \left\langle \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \right\rangle + \frac{2 + \eta_k L_{\pi_k^i}}{2 - \eta_k L_{\pi_k^i}} \eta_k^2 \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2. \end{aligned}$$

After rearranging terms, we have

$$\begin{aligned} \frac{2 - \eta_k L_{\pi_k^i}}{2} \eta_k B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) &\leq \|\mathbf{x}_k^i - \mathbf{x}_\star\|^2 - \|\mathbf{x}_k^{i+1} - \mathbf{x}_\star\|^2 + \frac{2 + \eta_k L_{\pi_k^i}}{2 - \eta_k L_{\pi_k^i}} \eta_k^2 \|\nabla f_{\pi_k^i}(\mathbf{x}_\star)\|^2 \\ &\quad - 2\eta_k \left\langle \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \right\rangle - \frac{2 - \eta_k L_{\pi_k^i}}{2} \eta_k B_{\pi_k^i}(\mathbf{x}_\star, \mathbf{x}_k^i). \end{aligned}$$

Therefore, we conclude the proof after applying

$$2 - \eta_k \hat{L} \leq 2 - \eta_k L_{\pi_k^i} \quad \text{and} \quad \frac{2 + \eta_k L_{\pi_k^i}}{2 - \eta_k L_{\pi_k^i}} \leq \frac{2 + \eta_k \hat{L}}{2 - \eta_k \hat{L}},$$

both of which hold by $L_{\pi_k^i} \leq \hat{L}$. □

D.2 Proof of Lemma 6

Proof. Note that $\nabla f(\mathbf{x}_\star) = \mathbf{0}$ now, which implies that

$$-\left\langle \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \right\rangle = \left\langle \nabla f(\mathbf{x}_\star) - \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \right\rangle.$$

Since $\mathbb{E} \left[\nabla f_{\pi_k^i}(\mathbf{x}_\star) \right] = \nabla f(\mathbf{x}_\star)$ under RR, after taking expectations on both sides, we obtain that

$$\mathbb{E} \left[-\left\langle \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \right\rangle \right] = \mathbb{E} \left[\left\langle \nabla f(\mathbf{x}_\star) - \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i \right\rangle \right]. \quad (61)$$

Now, we denote by $\ell_i(\mathbf{x}) \triangleq \langle \nabla f_i(\mathbf{x}_\star), \mathbf{x} \rangle$, $\forall i \in [n]$ and $\ell(\mathbf{x}) \triangleq \frac{1}{n} \sum_{i=1}^n \ell_i(\mathbf{x}) = \langle \nabla f(\mathbf{x}_\star), \mathbf{x} \rangle$. Then, (61) implies that

$$\mathbb{E} \left[-\left\langle \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \right\rangle \right] = \mathbb{E} \left[\ell(\mathbf{x}_k^i) - \ell_{\pi_k^i}(\mathbf{x}_k^i) \right] \stackrel{(a)}{=} \frac{1}{n} \sum_{j=1}^{i-1} \mathbb{E} \left[\ell_{\pi_k^i}(\mathbf{x}_k^i(i, j)) - \ell_{\pi_k^i}(\mathbf{x}_k^i) \right], \quad (62)$$

where (a) holds by Lemma 2.

Note that, for any $j < i$, we have

$$\begin{aligned} \ell_{\pi_k^i}(\mathbf{x}_k^i(i, j)) - \ell_{\pi_k^i}(\mathbf{x}_k^i) &= \left\langle \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i(i, j) - \mathbf{x}_k^i \right\rangle \leq \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\| \left\| \mathbf{x}_k^i(i, j) - \mathbf{x}_k^i \right\| \\ &\stackrel{(b)}{\leq} \eta_k \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\| \left\| \nabla f_{\pi_k^i}(\mathbf{x}_k^j) - \nabla f_{\pi_k^j}(\mathbf{x}_k^j) \right\|, \end{aligned}$$

where (b) follows from Lemma 3. Furthermore, we can bound

$$\left\| \nabla f_{\pi_k^i}(\mathbf{x}_k^j) - \nabla f_{\pi_k^j}(\mathbf{x}_k^j) \right\| \leq \left\| \nabla f_{\pi_k^i}(\mathbf{x}_k^j) - \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\| + \left\| \nabla f_{\pi_k^j}(\mathbf{x}_k^j) - \nabla f_{\pi_k^j}(\mathbf{x}_\star) \right\| + \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\| + \left\| \nabla f_{\pi_k^j}(\mathbf{x}_\star) \right\|,$$

which implies that

$$\begin{aligned} &\ell_{\pi_k^i}(\mathbf{x}_k^i(i, j)) - \ell_{\pi_k^i}(\mathbf{x}_k^i) \\ &\leq \eta_k \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\| \left\| \nabla f_{\pi_k^i}(\mathbf{x}_k^j) - \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\| + \eta_k \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\| \left\| \nabla f_{\pi_k^j}(\mathbf{x}_k^j) - \nabla f_{\pi_k^j}(\mathbf{x}_\star) \right\| \\ &\quad + \eta_k \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 + \eta_k \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\| \left\| \nabla f_{\pi_k^j}(\mathbf{x}_\star) \right\|. \end{aligned} \quad (63)$$

By AM-GM inequality, the following three inequalities hold

$$\begin{aligned} \eta_k \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\| \left\| \nabla f_{\pi_k^i}(\mathbf{x}_k^j) - \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\| &\leq \frac{4\eta_k^2 \hat{L}}{2 - \eta_k \hat{L}} \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 + \frac{2 - \eta_k \hat{L}}{16\hat{L}} \left\| \nabla f_{\pi_k^i}(\mathbf{x}_k^j) - \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2, \\ \eta_k \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\| \left\| \nabla f_{\pi_k^j}(\mathbf{x}_k^j) - \nabla f_{\pi_k^j}(\mathbf{x}_\star) \right\| &\leq \frac{4\eta_k^2 \hat{L}}{2 - \eta_k \hat{L}} \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 + \frac{2 - \eta_k \hat{L}}{16\hat{L}} \left\| \nabla f_{\pi_k^j}(\mathbf{x}_k^j) - \nabla f_{\pi_k^j}(\mathbf{x}_\star) \right\|^2, \\ \eta_k \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\| \left\| \nabla f_{\pi_k^j}(\mathbf{x}_\star) \right\| &\leq \frac{\eta_k}{2} \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 + \frac{\eta_k}{2} \left\| \nabla f_{\pi_k^j}(\mathbf{x}_\star) \right\|^2. \end{aligned}$$

Plug the above inequalities back into (63) to have

$$\begin{aligned}
\ell_{\pi_k^i}(\mathbf{x}_k^i(i, j)) - \ell_{\pi_k^i}(\mathbf{x}_k^i) &\leq \frac{2 - \eta_k \hat{L}}{16 \hat{L}} \left\| \nabla f_{\pi_k^i}(\mathbf{x}_k^j) - \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 + \frac{2 - \eta_k \hat{L}}{16 \hat{L}} \left\| \nabla f_{\pi_k^j}(\mathbf{x}_k^j) - \nabla f_{\pi_k^j}(\mathbf{x}_\star) \right\|^2 \\
&\quad + \left(\frac{8\eta_k \hat{L}}{2 - \eta_k \hat{L}} + \frac{3}{2} \right) \eta_k \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 + \frac{\eta_k}{2} \left\| \nabla f_{\pi_k^j}(\mathbf{x}_\star) \right\|^2 \\
&\leq \frac{2 - \eta_k \hat{L}}{8} B_{\pi_k^i}(\mathbf{x}_\star, \mathbf{x}_k^j) + \frac{2 - \eta_k \hat{L}}{8} B_{\pi_k^j}(\mathbf{x}_\star, \mathbf{x}_k^j) \\
&\quad + \left(\frac{8\eta_k \hat{L}}{2 - \eta_k \hat{L}} + \frac{3}{2} \right) \eta_k \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 + \frac{\eta_k}{2} \left\| \nabla f_{\pi_k^j}(\mathbf{x}_\star) \right\|^2,
\end{aligned} \tag{64}$$

where the last step is by Lemma 1, $L_{\pi_k^i} \leq \hat{L}$ and $L_{\pi_k^j} \leq \hat{L}$.

Combine (62) and (64) to obtain that

$$\begin{aligned}
\mathbb{E} \left[- \left\langle \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \right\rangle \right] &\leq \frac{1}{n} \sum_{j=1}^{i-1} \mathbb{E} \left[\frac{2 - \eta_k \hat{L}}{8} B_{\pi_k^i}(\mathbf{x}_\star, \mathbf{x}_k^j) + \frac{2 - \eta_k \hat{L}}{8} B_{\pi_k^j}(\mathbf{x}_\star, \mathbf{x}_k^j) \right] \\
&\quad + \frac{1}{n} \sum_{j=1}^{i-1} \mathbb{E} \left[\left(\frac{8\eta_k \hat{L}}{2 - \eta_k \hat{L}} + \frac{3}{2} \right) \eta_k \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 + \frac{\eta_k}{2} \left\| \nabla f_{\pi_k^j}(\mathbf{x}_\star) \right\|^2 \right].
\end{aligned}$$

We observe that $\mathbb{E} [B_{\pi_k^i}(\mathbf{x}_\star, \mathbf{x}_k^j)] = \mathbb{E} [B_{\pi_k^j}(\mathbf{x}_\star, \mathbf{x}_k^j)]$, $\forall j < i$ and $\mathbb{E} [\left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2] = \mathbb{E} [\left\| \nabla f_{\pi_k^j}(\mathbf{x}_\star) \right\|^2] = \sigma_\star^2$. Put everything together to conclude the desired inequality. $\square$

E Missing Proofs of Lemmas for Bound II

In this section, we provide the missing proofs of the lemmas presented in Subsection 4.3, which were used to prove the second convergence rate in Theorem 6.

Before presenting the proofs, we recall the notion introduced in (9). For any $k \in [K]$, the virtual sequence $\mathbf{y}_k^i, \forall i \in [n+1]$ follows the equations

$$\mathbf{y}_k^{i+1} = \mathbf{y}_k^i - \eta_k \nabla f_{\pi_k^i}(\mathbf{x}_\star), \forall i \in [n], \quad \mathbf{y}_{k+1}^1 = \mathbf{y}_k^{n+1}, \quad \text{where } \mathbf{y}_1^1 = \mathbf{x}_\star. \tag{65}$$

As shown in (10), the virtual sequence satisfies that

$$\mathbf{y}_k^{n+1} = \mathbf{y}_k^1 = \mathbf{x}_\star, \forall k \in [K]. \tag{66}$$

E.1 Proof of Lemma 7

Proof. Given $k \in [K]$ and $i \in [n]$, by the update rule of Algorithm 1 and the definition of $\mathbf{y}_k^{i+1}$ (see (65)), we have

$$\begin{aligned}
\left\| \mathbf{x}_k^{i+1} - \mathbf{y}_k^{i+1} \right\|^2 &= \left\| \mathbf{x}_k^i - \mathbf{y}_k^i - \eta_k \left(\nabla f_{\pi_k^i}(\mathbf{x}_k^i) - \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right) \right\|^2 \\
&= \left\| \mathbf{x}_k^i - \mathbf{y}_k^i \right\|^2 + \eta_k^2 \left\| \nabla f_{\pi_k^i}(\mathbf{x}_k^i) - \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 + 2\eta_k \left\langle \nabla f_{\pi_k^i}(\mathbf{x}_\star) - \nabla f_{\pi_k^i}(\mathbf{x}_k^i), \mathbf{x}_k^i - \mathbf{y}_k^i \right\rangle.
\end{aligned} \tag{67}$$

Note that, by Lemma 1, we can find that

$$\begin{aligned}
\left\| \nabla f_{\pi_k^i}(\mathbf{x}_k^i) - \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 &\leq L_{\pi_k^i} \left\langle \nabla f_{\pi_k^i}(\mathbf{x}_k^i) - \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \right\rangle \\
&= L_{\pi_k^i} \left\langle \nabla f_{\pi_k^i}(\mathbf{x}_k^i) - \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{y}_k^i - \mathbf{x}_\star \right\rangle + L_{\pi_k^i} \left\langle \nabla f_{\pi_k^i}(\mathbf{x}_k^i) - \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{y}_k^i \right\rangle \\
&\leq \frac{2 - \eta_k L_{\pi_k^i}}{4} \left\| \nabla f_{\pi_k^i}(\mathbf{x}_k^i) - \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 + \frac{L_{\pi_k^i}^2}{2 - \eta_k L_{\pi_k^i}} \left\| \mathbf{y}_k^i - \mathbf{x}_\star \right\|^2 + L_{\pi_k^i} \left\langle \nabla f_{\pi_k^i}(\mathbf{x}_k^i) - \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{y}_k^i \right\rangle,
\end{aligned}$$

where the last step is due to Cauchy-Schwarz inequality, AM-GM inequality, and $\eta_k < \frac{2}{\hat{L}} \Rightarrow 2 - \eta_k L_{\pi_k^i} > 0$. Rearrange terms in the above inequality to obtain that

$$\left\| \nabla f_{\pi_k^i}(\mathbf{x}_k^i) - \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 \leq \frac{4L_{\pi_k^i}^2}{4 - \eta_k^2 L_{\pi_k^i}^2} \|\mathbf{y}_k^i - \mathbf{x}_\star\|^2 + \frac{4L_{\pi_k^i}}{2 + \eta_k L_{\pi_k^i}} \left\langle \nabla f_{\pi_k^i}(\mathbf{x}_k^i) - \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{y}_k^i \right\rangle. \quad (68)$$

Next, we plug (68) back into (67) to have

$$\begin{aligned} \|\mathbf{x}_k^{i+1} - \mathbf{y}_k^{i+1}\|^2 &\leq \|\mathbf{x}_k^i - \mathbf{y}_k^i\|^2 + 2\eta_k \frac{2 - \eta_k L_{\pi_k^i}}{2 + \eta_k L_{\pi_k^i}} \left\langle \nabla f_{\pi_k^i}(\mathbf{x}_\star) - \nabla f_{\pi_k^i}(\mathbf{x}_k^i), \mathbf{x}_k^i - \mathbf{y}_k^i \right\rangle + \frac{4\eta_k^2 L_{\pi_k^i}^2}{4 - \eta_k^2 L_{\pi_k^i}^2} \|\mathbf{y}_k^i - \mathbf{x}_\star\|^2 \\ &= \|\mathbf{x}_k^i - \mathbf{y}_k^i\|^2 + 2\eta_k \frac{2 - \eta_k L_{\pi_k^i}}{2 + \eta_k L_{\pi_k^i}} \left(B_{\pi_k^i}(\mathbf{y}_k^i, \mathbf{x}_\star) - B_{\pi_k^i}(\mathbf{y}_k^i, \mathbf{x}_k^i) - B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) \right) + \frac{4\eta_k^2 L_{\pi_k^i}^2}{4 - \eta_k^2 L_{\pi_k^i}^2} \|\mathbf{y}_k^i - \mathbf{x}_\star\|^2 \\ &\stackrel{(a)}{\leq} \|\mathbf{x}_k^i - \mathbf{y}_k^i\|^2 + 2\eta_k \frac{2 - \eta_k L_{\pi_k^i}}{2 + \eta_k L_{\pi_k^i}} \left(B_{\pi_k^i}(\mathbf{y}_k^i, \mathbf{x}_\star) - B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) \right) + \frac{4\eta_k^2 L_{\pi_k^i}^2}{4 - \eta_k^2 L_{\pi_k^i}^2} \|\mathbf{y}_k^i - \mathbf{x}_\star\|^2 \\ &\stackrel{(b)}{\leq} \|\mathbf{x}_k^i - \mathbf{y}_k^i\|^2 - 2\eta_k \frac{2 - \eta_k L_{\pi_k^i}}{2 + \eta_k L_{\pi_k^i}} B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) + \frac{4 + \eta_k^2 L_{\pi_k^i}^2}{4 - \eta_k^2 L_{\pi_k^i}^2} \eta_k L_{\pi_k^i} \|\mathbf{y}_k^i - \mathbf{x}_\star\|^2, \end{aligned} \quad (69)$$

where we use $B_{\pi_k^i}(\mathbf{y}_k^i, \mathbf{x}_k^i) \geq 0$ in (a) and $B_{\pi_k^i}(\mathbf{y}_k^i, \mathbf{x}_\star) \leq \frac{L_{\pi_k^i}}{2} \|\mathbf{y}_k^i - \mathbf{x}_\star\|^2$ in (b).

Finally, we sum up (69) from $i = 1$ to n , use $\mathbf{x}_{k+1}^1 = \mathbf{x}_k^{n+1}$ and $\mathbf{y}_k^{n+1} = \mathbf{y}_k^1 = \mathbf{x}_\star$ (see (66)), and rearrange terms to obtain

$$2\eta_k \sum_{i=1}^n \frac{2 - \eta_k L_{\pi_k^i}}{2 + \eta_k L_{\pi_k^i}} B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) \leq \|\mathbf{x}_k^1 - \mathbf{x}_\star\|^2 - \|\mathbf{x}_{k+1}^1 - \mathbf{x}_\star\|^2 + \sum_{i=1}^n \frac{4 + \eta_k^2 L_{\pi_k^i}^2}{4 - \eta_k^2 L_{\pi_k^i}^2} \eta_k L_{\pi_k^i} \|\mathbf{y}_k^i - \mathbf{x}_\star\|^2.$$

Therefore, we conclude the proof after applying

$$\frac{2 - \eta_k \hat{L}}{2 + \eta_k \hat{L}} \leq \frac{2 - \eta_k L_{\pi_k^i}}{2 + \eta_k L_{\pi_k^i}} \quad \text{and} \quad \frac{4 + \eta_k^2 L_{\pi_k^i}^2}{4 - \eta_k^2 L_{\pi_k^i}^2} \leq \frac{4 + \eta_k^2 \hat{L}^2}{4 - \eta_k^2 \hat{L}^2},$$

both of which hold by $L_{\pi_k^i} \leq \hat{L}$. □

E.2 Proof of Lemma 8

Proof. Note that we have

$$L_{\pi_k^i} \|\mathbf{y}_k^i - \mathbf{x}_\star\|^2 \stackrel{(66)}{=} L_{\pi_k^i} \|\mathbf{y}_k^i - \mathbf{y}_k^1\|^2 \stackrel{(65)}{=} \eta_k^2 L_{\pi_k^i} \left\| \sum_{j=1}^{i-1} \nabla f_{\pi_k^j}(\mathbf{x}_\star) \right\|^2.$$

Therefore, we can bound

$$\mathbb{E} \left[\sum_{i=1}^n L_{\pi_k^i} \|\mathbf{y}_k^i - \mathbf{x}_\star\|^2 \right] = \eta_k^2 \mathbb{E} \left[\sum_{i=1}^n L_{\pi_k^i} \left\| \sum_{j=1}^{i-1} \nabla f_{\pi_k^j}(\mathbf{x}_\star) \right\|^2 \right] \leq \frac{\eta_k^2 n^2 \bar{L} \sigma_\star^2}{3},$$

where the last step is due to Lemma E.1 of Liu and Zhou (2024) (the constant here is slightly better, since $\nabla f(\mathbf{x}_\star) = \mathbf{0}$ in our setting leads to a provable improvement). □

F Rate for SGD

In this section, we present a convergence rate for standard SGD with epoch-dependent stepsizes. The proof relies on Lemma 5, a direct extension of Theorem 8 of Attia et al. (2025), as discussed previously.

Theorem 11. *Under Assumptions 1, 2, and 3, suppose $\eta_k < \frac{2}{\hat{L}}, \forall k \in [K]$, then SGD (Algorithm 1, with π_k^i sampled independently and uniformly from $[n]$) guarantees that*

$$\mathbb{E}[f(\bar{\mathbf{x}}_K) - f_\star] \leq \frac{2 \|\mathbf{x}_1^1 - \mathbf{x}_\star\|^2}{n \sum_{k=1}^K (2 - \eta_k \hat{L}) \eta_k} + \frac{8 \sum_{k=1}^K \eta_k^2 / (2 - \eta_k \hat{L})}{\sum_{k=1}^K (2 - \eta_k \hat{L}) \eta_k} \sigma_\star^2,$$

where $\bar{\mathbf{x}}_K$ is defined in (1).

Proof. We invoke Lemma 5 and drop the nonpositive term $-\frac{2-\eta_k \hat{L}}{2} \eta_k B_{\pi_k^i}(\mathbf{x}_\star, \mathbf{x}_k^i)$ to have, for any $k \in [K]$ and $i \in [n]$,

$$\frac{2 - \eta_k \hat{L}}{2} \eta_k B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) \leq \|\mathbf{x}_k^i - \mathbf{x}_\star\|^2 - \|\mathbf{x}_k^{i+1} - \mathbf{x}_\star\|^2 + \frac{2 + \eta_k \hat{L}}{2 - \eta_k \hat{L}} \eta_k^2 \left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 - 2\eta_k \left\langle \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \right\rangle. \quad (70)$$

Note that π_k^i is now sampled uniformly from $[n]$ and is independent from the history. Combining with the fact that $\nabla f(\mathbf{x}_\star) = \mathbf{0}$, this implies that

$$\mathbb{E} \left[B_{\pi_k^i}(\mathbf{x}_k^i, \mathbf{x}_\star) \right] = \mathbb{E} [f(\mathbf{x}_k^i) - f_\star], \quad \mathbb{E} \left[\left\| \nabla f_{\pi_k^i}(\mathbf{x}_\star) \right\|^2 \right] = \sigma_\star^2, \quad \mathbb{E} \left[\left\langle \nabla f_{\pi_k^i}(\mathbf{x}_\star), \mathbf{x}_k^i - \mathbf{x}_\star \right\rangle \right] = 0.$$

Therefore, after taking expectations on both sides of (70), multiplying both sides by 2, and using $\eta_k \hat{L} \leq 2$, we obtain that

$$(2 - \eta_k \hat{L}) \eta_k \mathbb{E} [f(\mathbf{x}_k^i) - f_\star] \leq 2\mathbb{E} [\|\mathbf{x}_k^i - \mathbf{x}_\star\|^2] - 2\mathbb{E} [\|\mathbf{x}_k^{i+1} - \mathbf{x}_\star\|^2] + \frac{8\eta_k^2}{2 - \eta_k \hat{L}} \sigma_\star^2.$$

Finally, a direct calculation yields the rate

$$\mathbb{E} [f(\bar{\mathbf{x}}_K) - f_\star] \leq \frac{2 \|\mathbf{x}_1^1 - \mathbf{x}_\star\|^2}{n \sum_{k=1}^K (2 - \eta_k \hat{L}) \eta_k} + \frac{8 \sum_{k=1}^K \eta_k^2 / (2 - \eta_k \hat{L})}{\sum_{k=1}^K (2 - \eta_k \hat{L}) \eta_k} \sigma_\star^2.$$

□